

Critical honeycomb chords with prescribed boundary endpoints

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Abstract

Critical self-avoiding walks between prescribed, macroscopically separated boundary ports of a regular honeycomb hexagon have length $R^{4/3+o(1)}$ in probability, where R is the scale of the hexagon. We prove the corresponding statements for half-plane arches, parallel cuts and nonparallel pure cuts. The half-plane law also has mean length $R^{4/3+o(1)}$. A separate strip argument gives endpoint mean laws on one density-one set of heights and in an aligned, critically weighted mixture of all even heights in a macroscopic interval.

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1 Introduction

A boundary-to-boundary walk can be weighted critically without fixing its length. In that law the separation of the endpoints supplies the spatial scale. Our question is whether prescribing the terminal boundary point, rather than summing over an interval, changes the power relating length to that scale. The distinction is substantial: a macroscopic terminal interval contains order R ports, and a bound on its total exceptional mass does not bound the exceptional mass at one specified port.

We use the regular honeycomb lattice, embedded as the centers of the triangles of an equilateral triangular tiling of side one. Neighboring centers are joined across a common triangle edge. A *port* is the midpoint of a boundary edge of a domain made of whole triangles. A path enters and leaves normally at its ports, visits only centers in the interior, and visits no center twice. Its length $L(\gamma)$ is the number of visited centers, and its critical weight is

$$w(\gamma) = \kappa^{L(\gamma)}, \quad \kappa = (2 \cos(\pi/8))^{-1} = (2 + \sqrt{2})^{-1/2}.$$

Ports carry no weight. For a domain D and boundary ports a, b , set

$$K_D(a, b) = \sum_{\gamma: a \rightarrow b \text{ in } D} w(\gamma).$$

Whenever the sum is positive and finite, dividing by it defines the critical chord law. The path has no other boundary exit. A *pure* cut follows one of the three tiling-edge directions; its admissible translates bound whole triangular rows. Distances refer to this fixed embedding.

Theorem 1.1 (Prescribed boundary endpoints). *Let D_R be convex pure lattice hexagons which, after scaling by R , converge to a regular hexagon. Let a_R, b_R be boundary ports whose scaled limits are distinct points in the relative interiors of sides. Then, for every $\varepsilon > 0$, the critical chord law satisfies*

$$\mathbb{P}_{D_R; a_R, b_R} \{R^{4/3-\varepsilon} \leq L(\gamma) \leq R^{4/3+\varepsilon}\} \longrightarrow 1.$$

The same conclusion holds in a full half-plane for prescribed boundary ports at distance R , in a strip of height comparable to R for one prescribed port on each of the two opposite boundary cuts, at

mutual distance $O(R)$, and in a wedge of angle $\pi/3$ or $2\pi/3$ formed by two nonparallel pure cuts, with one prescribed port on each ray. In the wedge, each endpoint has clearance comparable to R from the other cut and distance $O(R)$ from the apex. Constants are uniform on fixed compact sets of these nondegenerate scaled geometries. In each case the normalizing mass is at least $cR^{-5/4}$.

The theorem concerns each admissible choice of the two ports. A first step, proved in Section 3, fixes the source and sums the terminal port over a separated macroscopic interval. Its normalizing mass is of order $R^{-1/4}$. This intermediate law is also useful in its own right, and remains distinct from the prescribed-endpoint law.

For half-plane arches we additionally prove $\mathbb{E}L = R^{4/3+o(1)}$. This requires control of the weighted upper tail; convergence in probability alone would not imply it. We also record the terminal-summed mean law from the proof of [15, Theorem 8.1], where the gap ranges over $[R, 2R]$ and the arch diameter is confined to a sufficiently large fixed multiple of R . For strip bridges, a second argument proves a mean law uniformly over sublinear terminal offsets on a single set of heights of natural density one. It also proves the same mean exponent in the critical mixture of aligned bridges at all even heights in $[H, 2H]$. These mean statements retain their own height and weighting conventions.

Historical context. Nienhuis’s analysis of the dilute $O(n)$ model predicted the planar self-avoiding-walk exponents, including the length–distance power $4/3$ [11]. Duminil-Copin and Smirnov proved the honeycomb connective constant using a local winding relation and a boundary identity [3]. Their mid-edge convention is the port convention used here. For critical strip mass, Beaton, Bousquet-Mélou, de Gier, Duminil-Copin and Guttmann proved decay to zero in their work on surface adsorption [2]. Glazman and Manolescu gave a shorter proof and a logarithmic subsequence bound, and proved boundary two-point invariance for columnwise rhombic half-planes with angles in $[\pi/3, 2\pi/3]$ [6]. Krachun and Panagiotis subsequently proved a polynomial upper bound for strip-crossing mass and a quantitative sub-ballistic estimate for uniform honeycomb walks [9]. These results distinguish critical activity, decay of crossing mass, and length–distance exponents; the finite inputs below specify the stronger estimates used here.

The irreducible-bridge construction goes back to Kesten [8]; its honeycomb mid-edge form appears in [2, Appendix]. On the square lattice, the infinite critical law and its relation to strip-spanning laws are developed by Lawler–Schramm–Werner [10, Appendix] and Dyhr–Gilbert–Kennedy–Lawler–Passon [4, Section 2.4]. Gilbert’s appendix records Kennedy’s heuristic connecting strip mass of order $h^{-1/4}$ to an irreducible height tail of order $h^{-3/4}$ and renewal heights of order $n^{4/3}$ [5, Appendix]. Here whole irreducible bridges are independent increments; height, length and lateral displacement within one increment remain dependent.

The quantitative starting point is the sharp finite marked-polygon and bridge estimates of [16], together with the renewal estimates of [15]; Section 2 states exactly what we use. These are stronger inputs than the preceding critical and sub-ballistic results. The terminal-free fixed-height probability law and the prescribed half-plane arch probability law already appear in [15, Propositions 5.7 and 5.8]. We give a different proof of the latter, through common-level pair renewals and arch-kernel regularity, and develop the local estimates needed to assemble the prescribed-endpoint laws in the other geometries. The mean estimates require additional length-weighted control and retain separate finite inputs.

The three endpoint mechanisms. For two ports on one line, grow independent bridge sequences upward from the two ports and stop at their common renewal levels. Their common height transform has exponent $1/2$. A separate lateral window estimate gives $O(m^{-2})$ atoms for the difference after

m common blocks. This supplies both regularity of the arch kernel and control of a long final joining arch; the complete argument is in Section 4.

For parallel cuts one must match height and lateral displacement at once. A height-tail bound alone does not pay for one prescribed height. Section 5 bounds all fixed-order differences of the strip mass, starting from the finite projection representation in [16]. This local regularity gives the one-piece height atom bound in Section 6. Together with two-dimensional concentration and a sum of heightwise maximal atoms, it pays the parallel endpoint restriction in Section 7. For nonparallel cuts, small normal prefixes initially exhaust only a sum over terminal ports. A local visit estimate and an artificial-arc boundary identity then control the exceptional mass at the prescribed terminal port. Each case thus pays for its actual endpoint restriction before dividing by the chord mass; Section 8 completes the nonparallel case.

The means use two further arguments. In Section 9, the calibrated finite estimates of [12] and their renewal consequences control marked arms at both ends of an arch. The resulting length-weighted contributions are summable over arch heights at each prescribed gap. In Section 10, the finite strip moments of [14] feed a renewal-reward identity. A two-bridge boundary comparison controls variation between terminal ports after averaging over heights, leading to the density-one and aligned-mixture mean laws rather than an all-height pointwise claim.

2 Finite estimates and the irreducible law

We first identify the inputs that do not depend on prescribed endpoints. A strip of integral height h has physical thickness dh , where $d = \sqrt{3}/2$. Write B_h for its critical bridge mass from one fixed bottom port, summed over all compatible top ports, and put $B_0 = 1$. A strict bridge stays between its boundary cuts. Its terminal offset lies on the row-staggered lattice $\mathbb{Z} + h/2$ in horizontal side units. There is no translation factor in B_h .

Proposition 2.1 (Finite boundary and length inputs). *The following estimates hold in the port convention above.*

1. *For every integer $h \geq 0$, $B_h \asymp (1 + h)^{-1/4}$. For all sufficiently large integers h , the length-weighted bridge mass is at most $Ch^{13/12}$, and $B_h\{L \geq ch^{4/3}\} \geq ch^{-1/4}$. For every integer $h \geq 1$ and every $r \geq 0$, the mass of bridges of diameter exceeding r is at most $C(1 + r)^{-1/4}$.*
2. *In a convex pure domain, the mass from one boundary source of completed paths of diameter at least r is at most $C(1 + r)^{-1/4}$ for $r \geq 0$. For the half-plane arch kernel and integer $l \geq 1$, $K_l \asymp l^{-5/4}$ and $K_{l+1} \leq K_l$. The same monotonicity comparison is valid along one straight side of a convex pure truncation whenever both compared adjacent ports remain on that side. Arches of diameter at most Al retain mass $cl^{-5/4}$ for a fixed A .*
3. *From a prescribed port a with inward pure normal n , consider the terminal line $m \cdot (z - a) = s$, where m is its outward pure unit normal and the angle between n and m is at most $\pi/3$. Here $s > 0$ is the Euclidean distance from a to the terminal line. Crossings that stay in the initial half-plane and before the terminal line until their final exit have mass comparable to $s^{-1/4}$. For any fixed positive relative tolerance, and all sufficiently large admissible s with a threshold depending only on that tolerance, the lower bound can be confined within distance equal to that tolerance times s of the segment $a + [0, s]m$. The corresponding two-cut upper bound holds without confinement.*
4. *In a convex pure domain of diameter $O(R)$, the total length-weighted mass of chords, summed over ordered boundary endpoint pairs, is at most $R^{25/12+o(1)}$.*

These are the finite results of [16, Theorem 1.1, Propositions 7.1, 7.4, 10.1 and Corollary 8.1]. The statement records the input before use; the full analytic and finite geometric proofs are in that companion. In particular its free-boundary mean is a finite input, not a consequence of the endpoint theorem here. The disk-transfer construction [13, Theorem 1.1] provides an independent fixed-corridor first moment. We use the specified finite method at each application below; in particular the logarithmic confinement in the polynomial-vacuum route retains its own precision.

We use also the local contour identity with phase $e^{i\sigma W}$, $\sigma = 3/8$, where W is total turning. This is the phase $e^{-5iW/8}$ of the usual observable multiplied by the terminal tangent relative to the initial direction; see [3, Section 2, Lemma 1]. At an unvisited center the two outgoing weights sum to the incoming weight, since $2\kappa \cos(\pi/8) = 1$. Prefixes returning to a visited center cancel in pairs under reversal of their unfinished simple loop: their turn increments differ by $8\pi/3$, and hence their phases have opposite signs. Summing over a finite domain gives signed exit mass one. For a convex domain the real exit coefficients lie in $[\cos(3\pi/8), 1]$. Subtracting the identity in a convex subdomain bounds the mass of paths lost on restriction by a fixed multiple of the mass to the newly exposed sides. Exhaustion is justified by the diameter bound in Proposition 2.1. This is the local winding mechanism of [3]; below we explain the additional arc and spectator constructions when they are used.

An irreducible strict bridge has positive integral height and no intermediate singly crossed level. Every bridge has a unique ordered factorization into such pieces. The honeycomb mid-edge version of the irreducible construction is given in [2, Appendix]. Let $p(\gamma) = \kappa^{L(\gamma)}$ on irreducibles from a fixed port. Let H, Z, L, D denote respectively the height, transverse displacement, length and diameter of one piece.

Proposition 2.2 (Renewal inputs). *The mass p is a probability. Successive complete pieces have independent law p , and, for $s > 0$,*

$$\sum_{h \geq 0} B_h e^{-sh} = (1 - \mathbb{E}_p e^{-sH})^{-1} \asymp s^{-3/4} \quad (s \downarrow 0).$$

Moreover,

$$\begin{aligned} p(H > x) + p(D > x) &\leq Cx^{-3/4}, \quad \mathbb{E}_p[L; H \leq x] \leq x^{7/12+o(1)}, \\ p(L > n) &\leq n^{-9/16+o(1)}, \quad 1 - \mathbb{E}_p e^{-uL} = u^{9/16+o(1)}. \end{aligned}$$

Almost surely, for independent pieces, their first k heights and lengths sum respectively to $k^{4/3+o(1)}$ and $k^{16/9+o(1)}$, and their diameters sum to at most $k^{4/3+o(1)}$.

The proofs and the full joint-law conventions are [15, Propositions 2.2, 2.4, Remark 2.5 and Lemma 2.6]. Only the independent *pieces* are asserted here; their coordinates are not independent. B_h is a renewal occurrence mass, not a probability conditioned on a future height. No estimate in this proposition conditions a piece on its length.

Throughout, $f(R) = R^{a+o(1)}$ means $R^{a-\varepsilon} \leq f(R) \leq R^{a+\varepsilon}$ for every fixed $\varepsilon > 0$ at all sufficiently large scales, with fixed geometric parameters chosen first. A one-sided use retains just the corresponding inequality. All suprema of atoms are on their actual support lattice; off-lattice probabilities are zero.

3 A prescribed source and a terminal interval

Prescribing one endpoint and summing the other over a separated interval gives normalizing mass of order $R^{-1/4}$. We first establish this normalization by constructing interior connectors. We then prove the bridge and attachment estimates that transfer the length law to these chords. The attachment estimates will also be used for nonparallel cuts in Section 8.

3.1 Interior connectors

Proposition 3.1 (Interior connectors). *For two macroscopically separated side-interior ports in the hexagons of Theorem 1.1, the critical mass of connecting paths inside the hexagon is at least $cR^{-5/4}$. The same bound holds in a parallel strip of thickness comparable to R , with one port on each opposite cut and displacement $O(R)$, and in compact nondegenerate wedge geometries with one port on each ray, as in Theorem 1.1. The constants are uniform on fixed compact sets of these scaled configurations.*

Proof. We adapt the cut-averaging construction in the proof of [16, Proposition 10.2], placing the routes in the interior rather than the exterior. The geometric choices below ensure that the decomposition can be recovered for two different lists of cuts.

Routes and junction cuts. Work first in a hexagon and scale distances by R . Start with short segments in the inward normals at the two ports. Continue by disjoint simple arcs in the open interior to two parallel lanes in a free interior ball, preserving their cyclic order. The lanes approach one common pure transverse line from the same side. Their separation is a small fixed positive number, so a completing arch of diameter at most a fixed multiple of that separation fits in the free half-ball beyond the line. Keep that half-ball disjoint from all nonincident route portions.

Approximate the arcs by finitely many segments in pure normal directions, retaining the initial and terminal straight portions and making successive direction changes at most $\pi/3$. One can do this by alternating the two adjacent pure directions spanning each smooth tangent, with a short step in their common direction at a sector transition. A sufficiently fine fixed mesh preserves separation of nonlocal portions; on nearby portions, positive projection in a common direction preserves simplicity. Every segment has length a fixed positive multiple of R .

Index a branch's segments by their guiding directions n_j . The cut at the end of segment j has normal n_j , the *arriving* direction. The next leg starts with inward normal n_j and travels toward a cut with outward normal n_{j+1} . Proposition 2.1(iii) therefore keeps the incoming leg before the junction cut and the outgoing leg after it. Restrict each leg to a sufficiently narrow corridor about its segment. Each junction has its own small crossing neighborhood, avoided by all nonincident route portions. This is a local requirement: those portions need not avoid the entire infinite line of the cut.

The geometric choices have positive clearances. They persist in a neighborhood of the scaled endpoint data, including the small support perturbations allowed in Theorem 1.1. Compactness gives uniform choices of the number of segments and their relative tolerances.

One list and two lists of levels. Vary each junction cut in a fixed small window of admissible levels of width comparable to R . Choose the windows so that all corridors and stage orders just described persist for every list. The terminal level is shared by the two branches; all other levels vary independently. If there are k legs in total, the set Λ_R of level lists is thus a product of $k - 2$ ordinary windows and one shared terminal window. Give it the uniform product probability measure.

For a list λ , let $I_\lambda(\gamma)$ indicate that the chord splits into the confined legs and the completing arch at these cuts. The splitting is unique. At each isolated junction the two incident pieces meet the cut once and occupy its opposite half-planes; all other pieces avoid that crossing neighborhood. Put

$$F_R(\gamma) = \frac{1}{|\Lambda_R|} \sum_{\lambda \in \Lambda_R} I_\lambda(\gamma), \quad \int G d\mu = \sum_{\gamma: a \rightarrow b} \kappa^{L(\gamma)} G(\gamma).$$

Multiplication of the k point-to-line lower bounds and the confined arch lower bound gives

$$\int F_R d\mu \geq cR^{-k/4-5/4}.$$

For two lists, split each branch at both crossings in each junction neighborhood. The two crossings occur in their normal-level order: the incoming portions start below both levels, and the outgoing portions end above both. The intervening portion is consequently a strict bridge of the level difference. The portions between successive neighborhoods still have separation comparable to R and cost $O(R^{-1/4})$ each. There are k such long portions. Beyond the later shared terminal line the remaining arch costs $O(R^{-5/4})$.

For two independent uniform levels u, u' in a window of order R ,

$$\mathbb{E}(1 + |u - u'|)^{-\alpha} \leq C_\alpha R^{-\alpha} \quad (0 < \alpha < 1).$$

Every ordinary junction gives one difference bridge and uses this bound with $\alpha = 1/4$. The shared terminal level gives two difference bridges, one on each branch, and uses $\alpha = 1/2$. Ties are included by the convention $B_0 = 1$. Averaging the joint mass therefore gives

$$\int F_R^2 d\mu \leq C \underbrace{R^{-k/4}}_{\text{long portions}} \underbrace{R^{-(k-2)/4} R^{-1/2}}_{\text{difference bridges}} \underbrace{R^{-5/4}}_{\text{terminal arch}} = CR^{-2k/4-5/4}.$$

Cauchy–Schwarz now yields

$$\mu(F_R > 0) \geq \frac{(\int F_R d\mu)^2}{\int F_R^2 d\mu} \geq cR^{-5/4}.$$

For a wedge, add distant pure cuts at distance AR from the endpoints, with A fixed and sufficiently large. The endpoint clearances to the opposite supports permit initial inward segments of length cR . The resulting bounded convex patch contains an interior ball of radius $c'R$ accessible by the two separated routes above. After scaling, these choices persist on a neighborhood of each nondegenerate configuration; a finite cover of the compact family gives uniform clearances. The constructed paths stay in the full wedge. A parallel strip admits the same bounded-patch construction. This proves all the stated cases. $\square$

Theorem 3.2 (A source and a macroscopic terminal interval). *In the hexagons of Theorem 1.1, fix a source port tending after scaling to a side-interior point. Let the terminal ports form a macroscopic interval converging to a compact side-interior interval at positive distance from the source limit. Under critical weights, normalized over those chords, length is $R^{4/3+o(1)}$ in probability. The same holds when the source is summed over a separated macroscopic side-interior interval. For each fixed source the normalizer is comparable to $R^{-1/4}$.*

The diameter bound gives the upper normalization, and summing Proposition 3.1 over order- R terminal ports gives the lower one. We prepare the length estimates next; the proof of the theorem is completed at the end of this section.

3.2 Bridge length and displacement windows

Lemma 3.3 (Fixed-height bridge length). *Under critical weights on strict bridges of integral height h , with one initial port fixed and the terminal port summed, for every $\delta > 0$,*

$$\Pr\{h^{4/3-\delta} \leq L \leq h^{4/3+\delta}\} \longrightarrow 1.$$

The same conclusion holds after any restriction retaining a fixed positive fraction of B_h .

The fixed-height conclusion also appears in [15, Proposition 5.7]. We give the argument from the finite and renewal inputs selected in Section 2.

Proof. It suffices to treat $0 < \delta < 1$. The upper length tail follows from $B_h^{-1} \sum Lw \leq Ch^{4/3}$. For the lower tail, first obtain a height window under the one-piece law p . There are fixed $M, A > 1$ with

$$p\{x \leq H \leq Mx, D \leq Ax\} \geq cx^{-3/4} \quad (3.1)$$

for all sufficiently large x . Indeed $\mathbb{E}_p[1 - e^{-H/(Kx)}] \asymp (Kx)^{-3/4}$, whereas the contribution from $H \leq x$ is $O(x^{-3/4}/K)$ and that from $H > Mx$ is $O((Mx)^{-3/4})$. Choose first K , then M , sufficiently large. Finally the diameter tail permits $A \geq M$ large enough to retain the lower bound.

This window gives the concentration estimate

$$\sup_j \Pr(H_1 + \cdots + H_k = j) \leq Ck^{-4/3}. \quad (3.2)$$

To see this, symmetrize the characteristic function using an independent copy of H that is bounded with positive probability. In (3.1), take x to be a sufficiently small fixed multiple of $|\theta|^{-1}$. The cosine deficit on that event is bounded below, so the squared modulus loses at least $c|\theta|^{3/4}$ near zero. The modulus has only finitely many peaks on the lattice torus, repeats the same bound near each, and is bounded away from one on the remaining compact set. Fourier inversion proves (3.2).

For $k \geq 2$, split the pieces into comparable independent halves. If their total height is h , one half has height at least $h/2$, at cost at most $Ckh^{-3/4}$ by the tail and truncated first moment. The other half matches a height atom at cost $Ck^{-4/3}$. The case $k = 1$ costs at most $Ch^{-3/4}$. Hence, for fixed $\eta > 0$,

$$\sum_{1 \leq k \leq h^{3/4-\eta}} \Pr\left\{\sum_{j \leq k} H_j = h\right\} \leq Ch^{-3/4} \left(1 + \sum_{2 \leq k \leq h^{3/4-\eta}} k^{-1/3}\right) = o(B_h).$$

For the remaining counts put $t = h^{4/3-\delta}$. Exponential Markov and the length deficit give

$$\Pr\left\{\sum_{j \leq k} L_j < t\right\} \leq e(\mathbb{E}_p e^{-L/t})^k \leq e \exp\{-kt^{-9/16+o(1)}\}.$$

Choose $\eta < 9\delta/16$. Summing this geometric bound over $k > h^{3/4-\eta}$ is stretched-exponentially small. The renewal factorization proves the lower length tail after division by B_h . Dividing instead by any fixed positive fraction of B_h preserves both negligible exceptional masses. $\square$

We next arrange a displacement that points inward at both chord ends. This is needed when a translated chord is attached back to its original supports. Distances in the following geometric statement are physical; the bridge height h is still in row units.

Lemma 3.4 (Positive projections at two supports). *Let n, m be pure unit normals with $m \neq -n$. There are constants $b, B, A, c_0 > 0$ such that, for every fixed $\delta > 0$ and all sufficiently large integral h , the mass of strict n -bridges from a fixed port with endpoint displacement v , restricted by*

$$bh \leq m \cdot v \leq Bh, \quad D \leq Ah, \quad h^{4/3-\delta} \leq L \leq h^{4/3+\delta},$$

is at least $c_0 h^{-1/4}$. The constants may depend on n, m but not on δ ; the threshold may depend on δ .

Proof. If $m \cdot n > 0$, use a sufficiently narrow straight channel along n . The remaining case is an angle of $2\pi/3$. Let n' be the pure normal at angle $\pi/3$ to both, and put $T = dh$. First travel to an n' -normal cut at distance near $3T/2$ from the source, then travel in direction n to the top cut of the n -strip. Proposition 2.1(iii) guides the first leg along n' , even though its initial inward normal is n . At the intermediate position, both the n - and m -projections are near $3T/4$. The final leg advances about $T/4$ along n , leaving m -projection near $5T/8$. Small fixed tolerances keep both legs between the n -strip boundaries, on opposite sides of the intermediate cut, and within diameter $O(h)$.

Average that cut over a small window of order h . For one cut the two legs have product mass at least $ch^{-1/2}$. For two cuts, recovery in level order gives two long turned crossings and a difference bridge; the averaged joint mass is at most $Ch^{-3/4}$. Cauchy–Schwarz gives mass at least $ch^{-1/4}$ with the stated geometric restrictions. In both angular cases this is a fixed fraction of B_h . Lemma 3.3, used with slightly smaller exponent slack, removes the atypical lengths without changing the lower constant for all sufficiently large h . $\square$

3.3 The cost of two translation choices

Fix pure normals n, m with $m \neq -n$. For a lattice translation z with $n \cdot z > 0$, let $D_n(z)$ be the critical mass of strict n -bridges between the prescribed positions separated by z . For $n \cdot z < 0$, set $D_n(z) = D_n(-z)$; thus the bridge is always oriented in increasing n -level order. At zero height set $D_n(0) = 1$ and $D_n(z) = 0$ for nonzero z with $n \cdot z = 0$.

Lemma 3.5 (Mixed displacement estimate). *For every fixed $C_0 < \infty$ there is $C < \infty$ such that, for $s \geq 1$,*

$$\sum_{|z| \leq C_0 s} D_n(z)(1 + |m \cdot z|)^{-1/4} \leq Cs^{1/2}. \quad (3.3)$$

If Y is the m -projection of one n -directed irreducible displacement, then its independent sums also satisfy

$$\sup_y \Pr\left\{\sum_{j \leq k} Y_j = y\right\} \leq Ck^{-4/3}, \quad \sup_w \mathbb{E}\left(1 + \left|w + \sum_{j \leq k} Y_j\right|\right)^{-1/4} \leq Ck^{-1/3}. \quad (3.4)$$

Proof. The height-and-diameter window (3.1) and lateral reflection symmetry imply, for fixed $B > b > 0$,

$$p\{bx \leq |Y| \leq Bx\} \geq cx^{-3/4}.$$

Indeed the two reflected displacements have m -projections whose sum is $2(m \cdot n)dH$. Since $m \cdot n \neq 0$, at least one has magnitude comparable to H , while the diameter cutoff bounds both from above. The same symmetrization and Fourier argument as for (3.2) gives the first bound in (3.4) on the projection's actual lattice. For the second, sum this atom bound with weight $(1 + |y|)^{-1/4}$ over $|y| \leq k^{4/3}$, and bound the weight outside by $k^{-1/3}$. This is uniform over translated lattices and the shift w .

It suffices to sum over $n \cdot z \geq 0$ in (3.3). Use renewal factorization, splitting $k \geq 2$ pieces into comparable halves. The displacement restriction implies total height at most $C_1 s$. Height damping bounds this event on the first half by

$$C \exp\{-ck/s^{3/4}\}.$$

Conditioning on that half, the second bound in (3.4) applies to the other half, with a shift given by the first half's projection. Thus the sum is at most

$$C + C \sum_{k \geq 2} e^{-ck/s^{3/4}} k^{-1/3} \leq Cs^{1/2}.$$

The constant covers one piece and the zero-height match. No coordinates within one piece have been made independent. $\square$

3.4 Recoverable attachments

Consider the two inward support normals n, m at a source a and a terminal interval whose distances from a are comparable to R . When the supports differ, assume the endpoints have order- R clearance to the other support, as in the geometries above. We treat $m \neq -n$; opposite supports will use Lemma 3.3 directly. For identical supports their common half-plane replaces the two-cut domain.

Choose a small fixed $c > 0$ and a large scale $s \leq cR$. Translate an input chord by a lattice vector v with $|v| \leq Cs$ and both

$$g_1 = n \cdot v/d, \quad g_2 = m \cdot v/d$$

between fixed positive multiples of s . Attach a height- g_i bridge from each translated endpoint out to its original support. Restrict the two bridges to diameter at most $C's$ and length between $s^{4/3-\delta}$ and $s^{4/3+\delta}$. Such restrictions retain fixed fractions of the available bridge mass, by the preceding lemmas and the diameter tail. The branches stay in separated endpoint neighborhoods. They avoid the translated chord by its supports, and, for distinct supports, each stays beyond the translated support at the other end by endpoint clearance and the small choice of c .

Proposition 3.6 (Attachment moments). *Let $\mathcal{E}$ be an event of input chords from a to the stated terminal interval, of critical mass E . The fixed-source attachment construction just described has a translation count N_s on output chords and critical output measure μ_a such that*

$$\int N_s d\mu_a \geq cs^{1/2}E, \quad \int N_s^2 d\mu_a \leq CsR^{-1/4}. \quad (3.5)$$

Here the side-1 outer landing is exactly a ; the side-2 outer landing is free on its support. The estimate applies also to input chords in the full two-cut domain, without a diameter bound.

If the input chords lie in a convex pure domain of diameter $O(R)$, there is also a construction at scale $s = cR$ with both outer landings free, all outputs in one convex pure domain of diameter $O(R)$, and

$$\int N_R^{\text{free}} d\mu_{\text{free}} \geq cR^{3/2}E, \quad \int (N_R^{\text{free}})^2 d\mu_{\text{free}} \leq CR^{7/4}. \quad (3.6)$$

The measure μ_{free} sums critical weights over ordered output endpoint pairs. In both constructions output length is the input length plus the two attached lengths. Constants are uniform for fixed compact endpoint geometries and independent of the fixed length slack δ ; the required lower threshold for s may depend on δ .

Proof. For fixed source a , reverse the side-1 bridge: it starts at a and must have displacement exactly v . Let $\mathcal{V}_s$ be all translations in the size and positive-projection ranges above, with ranges wide enough to include those in Lemma 3.4. At each of order s heights, that lemma supplies restricted bridge mass at least $c_0s^{-1/4}$ after summing over its allowed displacements. Thus the total side-1 mass summed over v is at least $cs^{3/4}$. Side 2 has its outer landing free and supplies at least $cs^{-1/4}$ for each v .

Let $\mathcal{A}_v(\mathcal{E})$ be the set of output chords obtained with translation v , and define

$$N_s(\Gamma) = \sum_{v \in \mathcal{V}_s} \mathbf{1}_{\{\Gamma \in \mathcal{A}_v(\mathcal{E})\}}.$$

For a fixed v the output determines its decomposition: the crossings of the two translated supports separate the outer bridges from the central translated chord. On identical supports there are two crossings, localized near the two respective endpoints. Critical weights multiply, since the joining ports have no weight. This proves the first inequality in (3.5).

Recovering two translations. For v, v' , order the two crossings on each branch by that branch's inward normal height. Their orders may disagree between the two ends. Nevertheless both side-1 crossings occur before either side-2 crossing along the output chord: the outer prefixes and suffixes are localized in the two disjoint endpoint neighborhoods. Split into five portions: the side-1 outer bridge, its difference bridge, a central chord, the side-2 difference bridge, and the side-2 outer bridge.

The central portion is between the two innermost crossings. It stays inside both corresponding supports, because each construction remains beyond its own translated support after crossing, except for its localized branch at the other end when the supports coincide. Those other-end branches lie outside the central portion. Its initial point is prescribed once the two translations are fixed, its terminal point is free, and their separation is of order R . Dropping all other conditions bounds its mass by $CR^{-1/4}$.

Put $z = v' - v$. The side-1 difference has prescribed displacement z in one of its two orientations and costs $D_n(z)$. On side 2 the outer crossing is free, so the difference costs at most $C(1 + |m \cdot z|)^{-1/4}$. If the side-1 heights tie, its unique local crossing forces $v = v'$; thus the zero-height convention for D_n is the correct one. The side-2 outer bridge costs $Cs^{-1/4}$. Finally, if v_{out} is the earlier side-1 translation, the side-1 outer bridge costs $D_n(v_{\text{out}})$, with

$$\sum_{v_{\text{out}} \in \mathcal{V}_s} D_n(v_{\text{out}}) \leq C \sum_{h \leq Cs} B_h \leq Cs^{3/4}.$$

These upper bounds may be multiplied after dropping avoidance constraints between the five portions. Lemma 3.5 sums the two difference factors. The complete accounting is

$$\int N_s^2 d\mu_a \leq C \underbrace{s^{3/4}}_{\text{side-1 outer sum}} \underbrace{s^{1/2}}_{\text{two differences}} \underbrace{R^{-1/4}}_{\text{central chord}} \underbrace{s^{-1/4}}_{\text{side-2 outer}} = CsR^{-1/4}.$$

The argument used the two supports and endpoint clearance, not a bound on the diameter of the input chord.

Both outer landings free. Take $s = cR$ and sum over order R^2 translations in a fixed positive sector. Both outer bridges now have one free endpoint, so the first moment is at least $cR^2R^{-1/2}E$. The same two-list recovery keeps the inner side-1 crossing prescribed. Its outer bridge is now summed over its original-support landing, replacing the previous outer sum by $O(R^2R^{-1/4})$. The other three factors are unchanged. Thus

$$\int (N_R^{\text{free}})^2 d\mu_{\text{free}} \leq CR^{2-1/4}R^{1/2}R^{-1/4}R^{-1/4} = CR^{7/4}.$$

The localized attachments and $O(R)$ translations of bounded input chords all fit in one convex pure truncation of diameter $O(R)$. This proves (3.6). $\square$

3.5 Proof of the interval length law

Proof of Theorem 3.2. It suffices to prove the length bounds for $0 < \epsilon < 1$. The normalizing mass is comparable to $R^{-1/4}$, as shown above. If the two inward normals are opposite, the chords are a restriction of height- h bridges with $h \asymp R$, retaining a fixed fraction of B_h . Lemma 3.3 gives the result. Otherwise use Proposition 3.6.

For an input event of mass $E = p_0R^{-1/4}$, its fixed-source outputs have mass at least

$$\mu_a(N_s > 0) \geq \frac{(\int N_s d\mu_a)^2}{\int N_s^2 d\mu_a} \geq cp_0^2R^{-1/4}.$$

All these outputs have endpoints separated by order R , so their total mass is at most $CR^{-1/4}$ by the diameter bound.

Apply this to input lengths below $R^{4/3-\epsilon}$. Choose any fixed number M of distinct exponents γ_j in $(1-\epsilon/3, 1)$ and use $s_j = R^{\gamma_j}$. Choose $\delta > 0$ small enough that the M output length windows are disjoint and each attached length dominates the original short input. The lower mass constant is independent of this fixed δ . Consequently $Mcp_0^2 \leq C + o(1)$. Let $R \rightarrow \infty$ first and then $M \rightarrow \infty$. This proves $p_0 \rightarrow 0$ for the short-length event.

For input lengths above $R^{4/3+\epsilon}$, use both free landings. Equation (3.6) gives output mass at least

$$c \frac{R^3 E^2}{R^{7/4}} = cp_0^2 R^{3/4}.$$

Every output remains long. Proposition 2.1(iv) bounds its mass in the encompassing convex domain by

$$R^{-4/3-\epsilon} R^{25/12+o(1)} = o(R^{3/4}).$$

Thus $p_0 \rightarrow 0$ also for the long-length event. All bounds are uniform over the permitted source interval. Summing exceptional masses and normalizers over that interval proves the source-summed assertion. $\square$

4 Two prescribed ports on one line

Fix two boundary ports at distance l in a half-plane. The following argument grows independent prefixes from them, then joins those prefixes above their next common renewal level. The joining kernel must become insensitive to a sublinear change of its endpoint separation. We prove that regularity here, using the same pair process that controls the length of the joining path. The half-plane probability conclusion below is also established in [15, Proposition 5.8]. Here the common-level argument supplies local regularity of the arch kernel as well as the length law, which we then transfer to the same-side hexagon law.

Theorem 4.1 (Same-side probability law). *Under the critical half-plane law $w(\gamma)/K_l$, for every $\epsilon > 0$,*

$$\mathbb{P}\{l^{4/3-\epsilon} \leq L(\gamma) \leq l^{4/3+\epsilon}\} \longrightarrow 1.$$

The same conclusion holds in the hexagons of Theorem 1.1 when both scaled endpoints converge to distinct points in one side interior.

4.1 Common blocks and lateral concentration

We use the intersection-renewal construction discussed in [1, Section 1.1], proving the required identities directly. Distances along the base are physical distances, with port spacing one; heights are in row units. Run two independent sequences of upward irreducibles of law p , translating each new piece to the current endpoint. Stop at their next common positive renewal height. Let r be the resulting mass on finite pairs of terminating sequences, initially possibly a subprobability, and let T be their common height.

A pair block contains one complete bridge segment from each sequence, possibly made of several irreducibles. Let D_* be the maximum of the diameters of these two complete segments, and let Y be the second segment's lateral displacement minus the first's. In particular, $|Y| \leq 2D_*$. Successive common blocks regenerate independently: testing a prescribed pair of terminating sequences uses only the pieces up to their termination.

Lemma 4.2 (Common-block estimates). *The pair-block mass r is a probability. For $x \geq 1$,*

$$r(T > x) + r(D_* > x) \leq Cx^{-1/2}.$$

If $Y^{(m)}$ is the sum of the lateral differences of m independent pair blocks, then

$$\sup_j \Pr(Y^{(m)} = j) \leq Cm^{-2} \quad (m \geq 1).$$

Proof. The common renewal masses are B_h^2 , so

$$C(s) = \sum_{h \geq 0} e^{-sh} B_h^2 = (1 - \hat{r}(s))^{-1} \asymp s^{-1/2}, \quad \hat{r}(s) = \sum r e^{-sT}.$$

Sending s to zero shows that r has total mass one, and $1 - \hat{r}(s) \asymp s^{1/2}$ gives the height-tail estimate. For the diameter estimate use $s = D/x$ with D large fixed in $C(s)$. The fraction containing a pair-block with $D_* > x$ is at most $Cx^{-1/4}s^{-3/4}/C(s) < 1/2$, using the diameter estimate on one total bridge and the ordinary B_h bound on the other. More explicitly put $d_s = 1 - \hat{r}(s)$ and $q_s = \mathbb{E}_r[e^{-sT}; D_* > x]$. The fraction of terminated lists containing such a block is $q_s/(d_s + q_s)$, so the preceding bound below $1/2$ gives $q_s \leq d_s = O(x^{-1/2})$; remove the discount below height x , and above use the height tail from $1 - \hat{r}(s)$.

A lateral fluctuation in one common block. The common-height transform identifies the time scale. We next show that the transverse difference fluctuates on the same scale; otherwise it could not smooth a prescribed separation. We claim there are fixed $D_1 > D_0 > 0$ with

$$r(D_0x \leq |Y| \leq D_1x) \geq cx^{-1/2}$$

for large x . At every large height h , the ordinary bridge kernel has mass $\gtrsim h^{-1/4}$ of lateral displacements between two positive fixed multiples of h . For clarity, using physical thickness dh , go along the direction inclined at 60° to vertical to a pure tilted cut normal to that direction at distance near $(3/2)dh$ from the start, then vertically to the top, in small tolerances by the turned crossing estimate in Proposition 2.1(iii). The two legs lie on opposite sides of their joining cut. The first stays below the top, the second above the base, so they concatenate without mutual contacts and cross that cut exactly once. Average the concatenation indicator over a small fixed relative window of tilted levels. Its first moment is at least $ch^{-1/2}$. For two choices t, t' in row units, splitting at both tilted cuts gives two outer crossings and the intervening difference bridge. Their joint cost is at most

$$Ch^{-1/2}(1 + |t - t'|)^{-1/4}.$$

The outer crossings are still paid by the two-cut upper estimate. Averaging the gap factor over the order- h window gives a second moment of order at most $h^{-3/4}$. Cauchy–Schwarz therefore gives the asserted mass of off-axis bridges.

Normalize $C(1/x)$ to probability on terminated sequences. In heights $x \leq h \leq 2x$, applying these displacement restrictions in opposite directions retains a positive probability of total absolute displacement difference at least ax for some $a > 0$. On the other hand the number of pair-blocks there (under the normalized sum over all heights) is geometric on $0, 1, \dots$, of mean $O(x^{1/2})$, and the independent block laws conditional on that number are tilted by $e^{-T/x}/\hat{r}(1/x)$. By $|Y| \leq 2D_*$ the probability of any term with $|Y| > D_1x$ is $O(D_1^{-1/2})$, and the expectation of the absolute-value sum over terms with $|Y| < D_0x$ is $O(\sqrt{D_0}x)$. Taking the window sufficiently wide, we must thus see

a term within it with positive probability. The mean-count bound and tilt then imply the claim under r .

Y has a symmetric law, on the integers since the endpoints have the same row stagger. Consequently $Y^{(m)}$, the sum over m independent pair-blocks, has

$$\sup_j \Pr(Y^{(m)} = j) \leq Cm^{-2}.$$

Indeed symmetrizing with an independent copy (bounded with positive probability), the window bound yields $1 - |\mathbb{E}e^{i\theta Y}|^2 \gtrsim |\theta|^{1/2}$ near zero by taking x a sufficiently small fixed multiple of $1/|\theta|$. Peaks repeat the modulus at zero and there are only finitely many on the torus since the support has at least two integers. Integrate the modulus to the m -th power. $\square$

4.2 Regularity of the joining kernel

We next attach short prefixes and recover them from the completed arch. Figure 1 shows this construction.

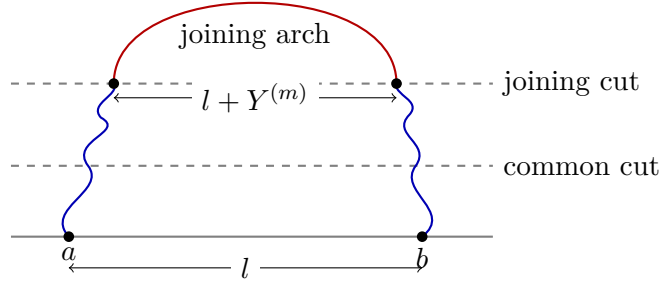


Figure 1: Two independent prefixes meet a common renewal height and are joined above it. In the restricted small-prefix construction, the prefixes remain near their respective starting ports. The first m levels with exactly two crossings recover the sampled pair blocks. This is a schematic: the lattice paths need not be monotone between these marked levels.

Lemma 4.3 (Sublinear changes of the gap). *For positive integers l, l' with $l \rightarrow \infty$ and $l'/l \rightarrow 1$,*

$$K_{l'}/K_l \rightarrow 1.$$

Proof. For initial separation j , call prefixes of m pair-blocks good if $\sum_{i \leq m} D_{*,i} < j/4$, of failure probability $O(m/\sqrt{j})$ by truncation. They are disjoint then, and we may join above the attained common level by any half-plane arch joining the endpoints. This gives

$$K_j \geq \mathbb{E}[\mathbf{1}_{\text{good}} K_{j+Y^{(m)}}].$$

Indeed product probabilities on the pair sequences are ordinary critical weights. The first m positive base-parallel levels having exactly two crossings of the output are precisely the m common renewals used: before reaching the joining level each branch must cross each level, and crossing just once is exactly renewal within that branch. The joining level itself has the two prescribed crossings. Thus the construction is injective.

Excluding a jump on a sublinear scale. Use monotonicity and the two-sided bounds of order $l^{-5/4}$. If the conclusion failed, along a sequence there would be drops at least $\epsilon l^{-5/4}$ between $l - w$ and $l + w$ with $1 \leq w = o(l)$. For large integer scales q set $m = \lfloor \sqrt{q} \rfloor$; then $\Pr(Y^{(m)} \leq -bq) \geq 1/4$ with some small fixed $b > 0$, by Lemma 4.2 and symmetry. Also $\Pr(|Y^{(m)}| > Aq) = O(A^{-1/2})$ by the pair diameter tail and truncation. Fix A large. Choose $w \ll q \ll l$ so the variation on $[l + bq/4, l + (A + b + 1)q]$ is $o(l^{-5/4})$; arbitrarily many disjoint such bands at geometric scales are available between these two scales, and their variations sum to $O(l^{-5/4})$. At $j = l + \lfloor bq/2 \rfloor$, on good prefixes with $|Y^{(m)}| \leq Aq$ the arch factor on the right of the inequality is at least $K_j - o(l^{-5/4})$. On those also having $Y^{(m)} \leq -bq$, the new gap lies below $l - w$, so the factor gains at least $\epsilon l^{-5/4}$ over K_j . Good has probability tending to one since $m/\sqrt{j} = O(\sqrt{q/l})$. The discarded tail loses at most $O(A^{-1/2})K_j$, whereas the gain occurs with probability at least $1/4 - O(A^{-1/2}) - o(1)$. Taking A sufficiently large in terms of ϵ contradicts the construction inequality. $\square$

4.3 Concentration under the prescribed-gap law

Proof of Theorem 4.1. Lemma 4.3 shows that sublinear perturbations preserve the joining mass. It remains to transfer the independent-prefix law and bound an unusually long joining arch. Take $m = \lfloor l^{1/2-\eta} \rfloor$ for small fixed $\eta > 0$. The good construction exhausts $1 - o(1)$ of total exact arch mass, since $\sum D_{*,i}/l \rightarrow 0$ in probability and the joining factor relative to K_l tends to 1 in probability there (and stays bounded on good prefixes). In the independent pair process the height after m grows as $m^{2+o(1)}$ almost surely by the height transform and the power-sum argument of Proposition 2.2. Each of the two individual cumulative lengths there grows as $(m^2)^{4/3+o(1)}$ by the almost sure individual irreducible-sequence laws, valid also along this subsequence of their renewals. Thus both prefixes' length exponents in l are arbitrarily close to $4/3$ with probability tending to one by taking η small, which transfers to good constructed paths by bounded reweighting.

Finally bound the contribution among these paths with joining length above $\frac{1}{2}l^{4/3+\epsilon}$. By the atom estimate and translating the first joining endpoint to a fixed origin it is at most

$$Cm^{-2} \left(CR^{-1/4} + l^{-4/3-\epsilon} R^{13/12+o(1)} \right), \quad R = l^{1+\gamma}.$$

Indeed after summing over the other joining endpoint this uses the half-plane diameter bound above R ; for smaller diameters the summed length mass with this fixed start is bounded by translating to order R equivalent starts along the base of a common encompassing convex truncation of size $O(R)$ and using the free-endpoints length bound $R^{25/12+o(1)}$ by occupancy. Choose $\gamma > 0$ sufficiently small relative to ϵ and then $\eta > 0$ sufficiently small that

$$2\eta < \gamma/4, \quad 2\eta + 13\gamma/12 < \epsilon.$$

Both displayed contributions are then $o(l^{-5/4})$. This proves the half-plane length concentration.

For the hexagon, the prescribed same-side separation satisfies $l \asymp R$. Proposition 3.1 gives normalizing mass at least $cl^{-5/4}$ inside the hexagon. The absolute exceptional mass there is at most the full half-plane exceptional mass, which is $o(l^{-5/4})$. Dividing by the finite-domain normalizer transfers both length tails. $\square$

5 Regularity of the strip partition function

The renewal identity alone gives a tail estimate for the height of one irreducible bridge. Prescribing the terminal port requires an estimate for a single height. Our first step is to show that the strip

partition function varies smoothly on its natural scale: every fixed forward difference gains the corresponding power of the strip width.

Proposition 5.1. *Let B_N be the critical mass of strict bridges from one fixed port to all ports on the opposite pure cut at integral height N . With $\Delta f_N = f_{N+1} - f_N$, for every fixed integer $\ell \geq 0$ and every $\eta > 0$ there is $C_{\ell,\eta} < \infty$ such that*

$$|\Delta^\ell B_N| \leq C_{\ell,\eta} N^{-1/4-\ell+\eta} \quad (N \geq 1).$$

The probabilistic application needs precisely Proposition 5.1: its passage to an exact height atom begins in Section 6. We now prove the proposition from the exact finite strip representation and its positive interpolation spaces. We state this input before introducing the holomorphic kernels used to estimate it. The spectral parameter $\lambda = \pi/8$ below is unrelated to a path length. There are three analytic tasks. We first transport the residual space to the disk and control its evaluation kernel, including the boundary column for the non-square-integrable input. We then recover the residuals at 0 and -4 from their Fourier transforms, retaining every fixed difference in N . Finally a separate positivity argument bounds the denominator away from zero; only then do we estimate the quotient for B_N .

5.1 The finite strip input

Put $b = 3\lambda$, $C_t(x) = \cosh(t\sqrt{x})$, $\mathcal{D}(x) = 2C_\lambda(x) - 1$, and $a_* = \mathcal{D}(-4) = 2\cos(2\lambda) - 1 > 0$. Define the positive measure μ on $(0, \infty)$ by

$$d\mu(x) = \frac{z\mathcal{D}(z^2)}{\sinh(4\lambda z)} dz, \quad x = z^2, \quad z > 0. \quad (5.1)$$

For $N \geq 1$, set

$$\begin{aligned} M_N &= \text{span}\{1, x, \dots, x^{N-1}, \partial_t^j C_t|_{t=\lambda} : 0 \leq j < N\}, \\ c_1 &= C_{\lambda/2}, \quad c_3 = C_{3\lambda/2}, \quad h_0 = \mathcal{D}^{-1}, \quad h_1 = c_1/\mathcal{D}, \\ g &= \int_0^\lambda C_t dt, \quad F_{0,N} = M_N + \mathbb{R}c_1, \quad G_N = M_N + \mathbb{R}g. \end{aligned}$$

Let $\Pi_N h \in F_{0,N}$ be the unique oblique projection preserving all pairings against G_N in $L^2(\mu)$, and write $r_N h = h - \Pi_N h$. The projection is also defined for c_3 , whose pairings with G_N are finite although $c_3 \notin L^2(\mu)$.

We use the following finite facts from the strip analysis in [16, Theorem 2.1 and Lemmas 3.1–3.4].

1. The above projection exists. Whenever the denominator is nonzero,

$$B_N = 2a_* \left[r_N h_0(-4) - \frac{r_N(c_3 - h_1)(-4) r_N h_0(0)}{r_N(c_3 - h_1)(0)} \right]. \quad (5.2)$$

The corresponding undivided statement is that there is a scalar s_N such that $2B_N = 4a_*[r_N h_0(-4) + s_N r_N h_0(0)]$ and $r_N(c_3 - h_1)(-4) + s_N r_N(c_3 - h_1)(0) = 0$.

2. For $0 < t < \lambda$, put $G_{N,t} = M_N + \mathbb{R}C_t$. On ordered positive nodes $X = (x_1 < \dots < x_{2N+1})$, the product of evaluation determinants for $(F_{0,N}, G_{N,t})$, times a positive base weight at every node, defines a positive density whenever its moments are finite. Its mean interpolant is the oblique projection against $G_{N,t}$. If $e_0 c_3$ denotes the interpolation residual at zero, then $-e_0 c_3$ is positive and increasing in each ordered node. Multiplying the base weight by an increasing positive function increases its expectation. Replacing $G_{N,t}$ by G_N is a positive mixture of these densities.

3. Let $F_N = F_{0,N} + \mathbb{R}c_3$, $A_N = \{f \in F_N : f(0) = 0\}$, and $U_N = \{f \in M_N : f(0) = 0\}$. For either determinant-product density $(F_{0,N}, G_{N,t})$ or $(A_N, G_{N,t})$, expectations of $\prod_i r(x_i)$, with $r(x) = (x + \alpha)/(x + \beta)$ and fixed $\alpha, \beta > 0$, differ by bounded factors from the corresponding symmetric expectations on (M_N, M_N) or (U_N, U_N) . The factors are uniform in N, t and remain so after any fixed number of such rational changes of the base weight. This is the interlacing reduction in the proof of [16, Lemma 3.2]: its positive weight and exponential-domination hypotheses persist under each such change, and the bound uses only the range of $\log r$, not t . The two symmetric expectations themselves differ by a bounded factor. These statements follow from interlacing of the positive roots of the compressed multiplication forms; a codimension-one compression changes the logarithmic product by at most the range of $\log r$.
4. For the probability measure μ_c proportional to $dz/\cosh(cz)$, $x = z^2$, let $\alpha_c = \pi/(2c)$ and $w_t = \tan^2(\alpha_c t)$. Then

$$A_t(x) = \cos(\alpha_c t) C_t(x), \quad \int A_t A_s d\mu_c = \frac{1}{1 - w_t w_s}. \quad (5.3)$$

Thus $A_t = \sum_{j \geq 0} P_j(x) w_t^j$ generates orthonormal polynomials P_j of degree j . The identity and its differentiated forms hold when $|t| + |s| < c$.

All base weights below have the required exponential domination. In particular these inputs apply to μ , to $dW = z dz/\sinh(bz)$, and to $r^j W$ for fixed j and fixed positive rational function r of the indicated form. We will prove both the estimates and the nonvanishing needed to use (5.2).

5.2 Transforming the residual space

Set

$$v(\omega) = \tan \frac{\pi\omega}{2b}, \quad J(\omega) = v'(\omega), \quad Q(v) = \frac{v^2(v^2 - v(\lambda)^2)}{1 - v^2 v(\lambda)^2}.$$

The strip $|\operatorname{Re} \omega| < b/2$ maps bijectively to the unit disk. The rational function Q is an even finite Blaschke product; its zeros are 0 with multiplicity two and $\pm v(\lambda)$ with multiplicity one. For an even holomorphic function A on the disk, define

$$(T_N A)(\omega) = J(\omega) Q(v(\omega))^N A(v(\omega)).$$

The factor Q^N encodes exactly the orthogonality conditions against M_N .

Indeed, the cosine transform identifies $h \in M_N^\perp \subset L^2(\mu)$ with

$$\int h(x) C_\omega(x) d\mu(x) = T_N A(\omega). \quad (5.4)$$

The transform is even, has a zero of order $2N$ at zero, and zeros of order N at $\pm\lambda$. Dividing by JQ^N therefore gives an even holomorphic A on the disk. Conversely these zero conditions imply the required orthogonality.

Here is the Hilbert-space content of this identification. Let $w_\mu(z)$ and $w_W(z)$ be the even extensions of the z -densities of μ and W . The Fourier datum of the transform $F = T_N A$ is $f(z) = h(z^2) w_\mu(z)$, so

$$F(\omega) = \frac{1}{2} \int_{\mathbb{R}} f(z) e^{\omega z} dz, \quad \|h\|_{L^2(\mu)}^2 = \frac{1}{2} \int_{\mathbb{R}} \frac{|f(z)|^2}{w_\mu(z)} dz.$$

We compare norms while holding F , and hence f , fixed. In the comparison measure W , the same transform is represented by $h_W = h d\mu/dW$; its squared norm replaces w_μ in the last integral by w_W . Moving to a parallel line multiplies the Fourier datum by $e^{z \operatorname{Re} \omega}$. Integrating Plancherel's identity across the strip supplies the reciprocal weight $\sinh(bz)/z = 1/w_W(z)$. The conformal Jacobian J then gives the transported comparison norm

$$\mathcal{G}_N^0(A, A) = c_b \int_{|v| < 1} |Q(v)|^{2N} |A(v)|^2 d^2v, \quad c_b > 0. \quad (5.5)$$

For completeness, the converse Fourier representation follows by Fourier transforming the Cauchy–Riemann equation locally in frequency: the data on parallel lines must have the same exponential multiplier. Strip square integrability gives the stated weighted L^2 data, and division by the even weight gives h . Thus no unproved surjectivity of the transform is being used.

Write $\mathcal{G}_N(A, A) = \|h\|_{L^2(\mu)}^2$ for the norm in (5.4). The density ratio of μ to W is bounded above and below by positive constants, so $\mathcal{G}_N \asymp \mathcal{G}_N^0$ uniformly in N . More precisely, their difference is a fixed bounded sesquilinear form on the two traces $T_N A(\pm d_{\text{tr}} + iy)$, for a fixed $d_{\text{tr}} < b/2$ sufficiently close to $b/2$. To see this, the difference of the reciprocal Fourier weights is, up to a fixed factor,

$$z^{-1} \left\{ \frac{\sinh(4\lambda z)}{\mathcal{D}(z^2)} - \sinh(bz) \right\}.$$

It is bounded at zero and is $O(e^{2d_{\text{tr}}|z|})$ at infinity: the terms of largest exponential order cancel. Plancherel on the two parallel lines therefore bounds the difference form by their L^2 trace norms.

If $A(v) = \sum_{j \geq 0} a_j v^{2j}$, both transported norms are equivalent to

$$\|A\|_N^2 = \sum_{j \geq 0} \frac{|a_j|^2}{N + j}. \quad (5.6)$$

The boundary phase speed of the finite Blaschke product Q is smooth and strictly positive. Hence $|Q(re^{i\theta})|^{2N}$ is bounded above by $e^{-cN(1-r)}$, and bounded below by $e^{-CN(1-r)}$ for r near one. Angular orthogonality of the powers, followed by radial integration, proves the equivalence.

Every fixed ℓ th forward difference of $\mathcal{G}_N^0$ costs $O_\ell(N^{-\ell})$ as a form in (5.6), since it inserts $(|Q|^2 - 1)^\ell$ in the area integral. The same is true for $\mathcal{G}_N$. On each of the fixed interior traces, $v \rightarrow \pm i$ nontangentially at the two ends and $Q(\pm i) = 1$; thus $|1 - Q| \leq C(1 - |Q|)$. Each difference of T_N inserts $Q - 1$, and comparison with $T_{\lfloor N/2 \rfloor}$ costs $O(N^{-1})$. Interior trace norms are bounded by strip area norms, by the mean-value inequality for holomorphic functions. The discrete product rule therefore gives the same bound for the correction form. All norms with finitely shifted indices are equivalent. Differencing $\mathcal{G}_N^{-1} \mathcal{G}_N = I$ consequently shows that every ℓ th difference of the inverse form costs $O_\ell(N^{-\ell})$ as well.

5.3 The model evaluation kernel

Denote by $K_N^0(v, u)$ the evaluation kernel for $\mathcal{G}_N^0$, analytic in v and anti-analytic in u . Thus, with the inner product linear in its first argument, $A(u) = \mathcal{G}_N^0(A, K_N^0(\cdot, u))$ for every even function in the space and every $|u| < 1$. The following bounds are needed also at boundary points, provided the two variables do not meet on the even diagonal.

Lemma 5.2. *On every compact subset of the closed bidisk on which $|1 - v^2 \bar{u}^2| > 0$, the kernel has continuous boundary values and*

$$|\Delta^\ell K_N^0(v, u)| \leq C_\ell N^{1-\ell}, \quad \frac{K_N^0(v, u)}{N} \longrightarrow \frac{O(v) \overline{O(u)}}{1 - v^2 \bar{u}^2}.$$

Here O is even, holomorphic and nonvanishing on a neighborhood of the closed disk, and positive on both the real and imaginary diameters.

Proof. Let P_{jk} be the Gram matrix of v^{2j} in (5.5), and $R = P^{-1}$. Thus $K_N^0(v, u) = \sum_{j,k \geq 0} v^{2j} R_{jk} \bar{u}^{2k}$ in the open bidisk. Write δ for simultaneous forward difference in j, k . For all fixed nonnegative integers ℓ, s ,

$$|\Delta^\ell \delta^s P_{jk}| \leq C_{\ell,s} e^{-c|j-k|} (N+j+k)^{-1-\ell-s}. \quad (5.7)$$

In the radial integral the differences insert $(|Q|^2 - 1)^\ell (r^4 - 1)^s$. For the angular Fourier coefficient, shift θ by a sufficiently small fixed imaginary amount in the direction giving decay in $|j-k|$. The continued expression $Q(re^{i\theta})Q(re^{-i\theta})$ equals one at $r = 1$. Its positive radial logarithmic derivative stays positive under this small shift, and compactness away from the boundary gives a strict bound below one there. It therefore decays as $e^{-c(1-r)}$ and differs from one by $O(1-r)$. Radial integration proves (5.7).

The inverse matrix satisfies

$$|\Delta^\ell \delta^s R_{jk}| \leq C_{\ell,s} e^{-c|j-k|} (N+j+k) N^{-\ell} (1 + \min\{j, k\})^{-s}. \quad (5.8)$$

First take $\ell = s = 0$. We use the exponential-conjugation argument for localized inverses; compare Jaffard [7, Proposition 2]. The parameter differences will be retained explicitly below. Conjugating P by diagonal factors $\sqrt{N+j}$ gives a positive operator with bounded positive inverse, by (5.6). Conjugation of this normalized operator by weights of sufficiently small exponential logarithmic slope changes it by an arbitrarily small operator norm, by row and column sums in (5.7). Its inverse remains bounded. Truncate these weights and pass to the limit to obtain exponential decay of inverse entries, proving the undifferenced case.

Here are details for the differences. Write $(SA)_{jk} = A_{j+1,k+1}$, so $\delta A = SA - A$. The half-line matrix product has the exact shift rule

$$S(AB) = (SA)(SB) + E(A, B), \quad E(A, B)_{jk} = A_{j+1,0} B_{0,k+1}.$$

In particular $PR = I$ gives, with the N dependence explicit in the second identity,

$$\delta R = -R(\delta P)SR - RE(P, R), \quad \Delta R_N = -R_N(\Delta P_N)R_{N+1}.$$

The first identity displays the boundary contribution that is absent from a difference in N . Bounds of the form in (5.8), with $(N+j+k)^a$ in place of its first power, compose under matrix multiplication by adding the orders a ; convolution of exponential decays absorbs every fixed polynomial in index distance. The boundary term $E(P, R)$ and its fixed index shifts have exponential decay in $j+k$, which pays every displayed index-decay factor. First establish all pure N differences from the second identity. Then induct on s in the first identity, treating every fixed ℓ at each step. Discrete product rules use only lower-order index differences of R ; the extra difference on P and the boundary localization give the additional index decay. Together with (5.7), these bounds prove (5.8).

These estimates also give convergence at the boundary off the even diagonal. On each matrix diagonal $j-k$ fixed, put $n = \min\{j, k\}$ and use a smooth dyadic partition in n . The variable factor is $(v^2 \bar{u}^2)^n$. Summing by parts s times, and then summing the exponentially decaying matrix diagonals, bounds the block $n \asymp m$ of $N^{\ell-1} \Delta^\ell K_N^0$ by $C_s m^{2-s}$, uniformly whenever $|1 - v^2 \bar{u}^2|$ is bounded below. Choosing $s > 2$ gives uniform convergence of these dyadic blocks and the asserted difference bound and continuity. In the open bidisk this sum agrees with the absolutely convergent kernel, so it gives its boundary values.

For fixed j, k , boundary radial Laplace integration shows that NP_{jk} tends to a positive Toeplitz matrix. Its symbol is a positive constant times the reciprocal boundary phase speed of Q , restricted

to even powers. Exponential matrix decay allows passage to the limit in both inverse identities, so R_{jk}/N tends to the inverse of this Toeplitz form. Factor the positive symbol as $1/|O|^2$, by harmonically extending its logarithm and taking a harmonic conjugate, normalized by $O(0) > 0$. This is the usual outer factor construction; see Simon [17, Section 5]. Real analyticity, evenness, and reflection symmetry of the symbol give the asserted analytic continuation, nonvanishing, and symmetries of O . In particular O is real and cannot change sign on either diameter. Multiplication by $1/O$ is an isometry from the resulting weighted even Hardy space onto the ordinary even Hardy space. The latter has kernel $(1 - v^2 \bar{u}^2)^{-1}$. This identifies the limit; the same summable dyadic tail bound makes the convergence uniform on the stated compact sets. $\square$

The model has now isolated the only large part of the evaluation kernel. The remaining change of measure contributes a bounded vector, which is small enough for every residual estimate below.

Let K_N denote the kernel for $\mathcal{G}_N$. If $u = v(t)$, $0 \leq t \leq \lambda$, or $t \uparrow b/2$, then

$$\|\Delta^\ell(K_N(\cdot, u) - K_N^0(\cdot, u))\|_N \leq C_\ell N^{-\ell}. \quad (5.9)$$

At $t = b/2$, where $u = 1$, the difference of columns has a limit in $\|\cdot\|_N$. Neither full boundary column is asserted to be a finite-norm vector. We define $K_N(\cdot, 1)$ as the model boundary column plus this finite-norm correction, with evaluations in the open disk obtained as limits from inside. Indeed the pairs formed by a point on either fixed interior trace and these u are uniformly off the even diagonal. On the traces,

$$|J| \asymp e^{-\pi|y|/b}, \quad 1 - |Q| \gtrsim e^{-\pi|y|/b}.$$

Lemma 5.2 and discrete product rules therefore give

$$\|\Delta^\ell(T_N K_N^0(\cdot, u))\|_{L^2(\{\pm d_{\text{tr}} + iy\})} \leq C_\ell N^{-\ell}.$$

For example the undifferenced integrand is bounded by

$$C N e^{-\pi|y|/b} \exp\{-c N e^{-\pi|y|/b}\}.$$

Its squared integral is bounded; each difference supplies its stated additional factor. Apply the bounded correction form to this column and solve with $\mathcal{G}_N^{-1}$. The resolvent identity and the inverse difference bounds prove (5.9). Dominated convergence on the traces gives its endpoint interpretation.

Evaluation at a fixed interior disk point costs $O(\sqrt{N})$ in (5.6). Consequently, for v, u corresponding to $0 \leq t \leq \lambda$, the actual kernel has the same strictly positive leading limit after division by N , with ℓ th differences $O(N^{1-\ell})$. The same holds when exactly one variable is 1.

5.4 Residual estimates at the two evaluation points

We prove the following bounds, which are the full analytic input to the quotient in (5.2):

$$\begin{aligned} |\Delta^\ell r_N h_j(-a)| &\leq N^{-1+b\sqrt{a}/\pi-\ell+o(1)} \quad (j = 0, 1), \\ |\Delta^\ell r_N c_3(-a)| &\leq N^{b\sqrt{a}/\pi-\ell+o(1)} \quad (a = 0, 4), \\ -r_N c_3(0) &\geq c > 0. \end{aligned} \quad (5.10)$$

All orders ℓ are fixed. The estimates with $o(1)$ mean that each fixed positive exponent slack is allowed, with its own constant.

First take the orthogonal residual against M_N alone. For input $h = h_j$ its transform is $T_N A_{h,N}$, where $A_{h,N}$ represents the functional obtained by pairing h with the residual space. Write

$$h(z^2) = \int_{\mathbb{R}} f_h(y) \cos(yz) dy.$$

The function f_h is integrable against $e^{u|y|}$ for each $u < \pi/(3\lambda) = \pi/b$. This follows by Fourier inversion and shifting within closed narrower strips: the first zeros of $\mathcal{D}(z^2)$ have imaginary part $\pi/(3\lambda)$, and the functions decay exponentially along every such horizontal line. Thus the functional on A is $\int f_h(y)(T_N A)(iy) dy$.

On $\|A\|_N \leq 1$, the area mean-value bound and comparison to $T_{\lfloor N/2 \rfloor}$ give

$$|(\Delta^\ell T_N A)(iy)| \leq C_\ell N^{-\ell} \exp\{-cN e^{-\pi|y|/b}\}.$$

Indeed $1 - |Q(v(iy))| \gtrsim e^{-\pi|y|/b}$ and $|1 - Q| \lesssim 1 - |Q|$ there. Integrating against f_h , then using the inverse-form differences, yields

$$\|\Delta^\ell A_{h,N}\|_N \leq N^{-1-\ell+o(1)}. \quad (5.11)$$

For an input C_s with $0 \leq s < b/2$, the representing vector is instead $J(s)Q(v(s))^N K_N(\cdot, v(s))$. As $s \uparrow b/2$, all finite projection pairings and the residual transforms on compact subsets of the strip converge. Thus the formula extends to c_3 using the boundary column just constructed and $Q(1) = 1$; it does not assert a bounded Hilbert-space evaluation functional at $u = 1$.

We now impose the one remaining oblique constraint. Put $u_0 = v(\lambda/2)$. The transform of $r_N h$ is T_N applied to

$$A_{h,N} - K_N(\cdot, u_0) \frac{\chi_N(A_{h,N})}{\chi_N(K_N(\cdot, u_0))}, \quad (5.12)$$

where χ_N averages values at $v(t)$, $0 \leq t \leq \lambda$, with probability proportional to $J(t)|Q(v(t))|^N dt$. To check the formula, orthogonality to M_N is already encoded in T_N , and $\int_0^\lambda C_t dt = g$. On this interval Q has constant sign away from its zeros, so this last condition is precisely $\chi_N = 0$.

Every fixed ℓ th difference of the probabilities defining χ_N is $O_\ell(N^{-\ell})$ in total variation. Divide their exponential base by its maximum. Differencing inserts powers of the gap to one. Integrals at exponents $N/2$ and N are comparable: the gap vanishes to finite order at its finitely many maxima, by analyticity, and the one-dimensional radial Laplace estimate applies in disjoint neighborhoods of them. Normalization and the quotient difference rule give the claim. Kernel positivity and its leading limit now show

$$\chi_N(K_N(\cdot, u_0)) \geq cN, \quad |\Delta^\ell \chi_N(K_N(\cdot, u_0))| \leq C_\ell N^{1-\ell}.$$

For h_j , the subtracted vector in (5.12) has the same bound as (5.11): use the $O(\sqrt{N})$ interior evaluation bound, the $O(\sqrt{N})$ column norm, and the denominator of order N , with their corresponding difference bounds. For c_3 , the scalar multiplier of the interior column has ℓ th differences $O_\ell(N^{-\ell})$. The model columns at 1 and u_0 have T_N -images on the imaginary axis bounded by $O(N^{-\ell})$ after ℓ differences, by the off-diagonal kernel estimate; their finite-norm corrections satisfy the same bound. We have proved, with $F_{h,N}(\omega) = \int (r_N h) C_\omega d\mu$, that

$$|\Delta^\ell F_{h,N}(iy)| \leq N^{a_h - \ell + o(1)}, \quad a_{h_0} = a_{h_1} = -1, \quad a_{c_3} = 0, \quad (5.13)$$

uniformly for $y \in \mathbb{R}$.

Growth on a shifted contour. The bound at -4 requires a contour argument; it is not an evaluation on the integration axis. For $h = h_0, h_1, c_3$, put

$$\rho_{h,N}(z) = (r_N h)(z^2) \frac{z \mathcal{D}(z^2)}{\sinh(4\lambda z)}.$$

We first prove that, for every fixed $Y > 0$ which is not an even integer,

$$\int_{\mathbb{R}} |\rho_{h,N}(z + iY)| dz \leq C_Y N^{bY/\pi + C}, \quad (5.14)$$

where the additive exponent C is independent of Y . This uniformity will allow us to choose Y after fixing the desired difference order. The factors $\mathcal{D}$ remove the poles of h_0, h_1 in this product.

In the orthogonal polynomial expansion (5.3) with $c = b$, coefficients of an element of $F_{0,N}$ beyond index $N - 1$ have the form

$$q^j p(j) + d_0 q_1^j, \quad \deg p < N, \quad 0 < q_1 = w_{\lambda/2} < q = w_\lambda < 1.$$

Divide by q^j and take N consecutive forward differences to bound d_0 by $e^{O(N)}$ on the unit sphere. Lagrange interpolation on N consecutive indices bounds the remaining polynomial. Consequently the coefficients decay exponentially beyond $C'N$, relative to the norm, for a fixed sufficiently large C' . Fix $2\lambda < c < b$ and repeat the coefficient-tail argument in the orthonormal polynomial basis for μ_c . For each $f \in F_{0,N}$, its truncation at degree $O(N)$ approximates f with exponentially small relative μ_c -norm error. The small-circle generating-function bound gives $|P_j^{(c)}(z^2)| \leq C_c^{j+1} e^{cz/4}$ for this basis. Choosing a fixed C' sufficiently large therefore makes the truncation's μ_c -norm on $z > C'N$ exponentially small relative to $\|f\|_{\mu_c}$. By the triangle inequality, a fixed positive fraction of the full μ_c -norm of f lies on $z \leq C'N$. On this interval the density ratio $d\mu_c/d\mu_b$ is at most $e^{O(N)}$, and hence

$$\|f\|_{\mu_c} \leq e^{O(N)} \|f\|_{\mu_b}, \quad f \in F_{0,N}.$$

All constants here may depend on the fixed c , not on N .

Pairing against c_3 with μ has norm at most a fixed power of N on this space. On $z \leq N^2$ use Cauchy-Schwarz; since $c_3(z^2)^2 d\mu/dz = O(1+z)$, its truncated squared integral is polynomial in N . Beyond N^2 use the just established μ_c comparison: the other factor in Cauchy-Schwarz decays as $e^{-(b-c)z}$, which dominates $e^{O(N)}$. Thus the orthogonal projection of c_3 on M_N has polynomially bounded norm. Its extra oblique projection term also does, by (5.12); for h_j this already follows from (5.11).

Finally the generating formula on $|w_t| = \exp(-1/(j+1))$ gives

$$|P_j((z + iY)^2)| \leq C_Y (j+1)^{1/2 + bY/\pi} e^{b|z|/2}.$$

Here the corresponding parameter satisfies $|\operatorname{Re} t| < b/2$ and $|\operatorname{Im} t| \leq (b/\pi) \log(j+1) + O(1)$. Summing with the coefficient bounds, and including the elementary direct inputs, proves (5.14): multiplication by the density leaves integrable exponential decay in $|z|$. In particular the exponent has coefficient bY/π and an additive constant independent of Y , as required.

Recovering the two evaluations. Fix the difference order ℓ and an arbitrarily small $\eta > 0$, and put $S_N = (b/\pi + \eta) \log N$. In the full even Fourier integral for $F_{h,N}(iy)$, shift the z contour to $\operatorname{Im} z = Y$. The only crossed poles are the simple poles of $1/\sinh(4\lambda z)$ at $z = 2in$, $n \geq 1$. The growth bound gives, for $y \geq S_N$,

$$F_{h,N}(iy) = \sum_{0 < 2n < Y} b_{n,N} e^{-2ny} + O_Y(e^{-Yy} N^{bY/\pi + C}), \quad (5.15)$$

where $b_{1,N}$ is a fixed nonzero multiple of $r_N h(-4)$. Apply (5.15) also at $N+1, \dots, N+\ell$, keeping the same S_N , and take differences. At $y = S_N + j$ for enough consecutive integers $j \geq 0$, the exponentials form a fixed Vandermonde system. Choose the non-even height $Y > 2$ sufficiently large depending on ℓ, η , and then let N tend to infinity. Since C is independent of Y , the remainder $N^{-\eta Y + C}$ is negligible at the required difference order. Equation (5.13) then bounds every coefficient $e^{-2nS_N} \Delta^\ell b_{n,N}$ by $N^{a_h - \ell + o(1)}$. For $n = 1$ this gives

$$|\Delta^\ell r_N h(-4)| \leq N^{a_h + 2b/\pi + 2\eta - \ell + o(1)}.$$

Letting η be arbitrarily small proves the claimed exponent. The same expansion bounds the integral over $y \geq S_N$ by $N^{a_h - \ell + o(1)}$. Below S_N , integrate (5.13); its logarithmic factor is absorbed in $o(1)$. Fourier inversion at $z = 0$, where the density of μ is positive and finite, gives the first two lines of (5.10) also at $a = 0$.

5.5 A uniform nonzero denominator

It remains to prove $-r_N c_3(0) \geq c$. An upper estimate for the residual would not suffice to divide by it. First replace G_N by $G_{N,t}$, $0 < t < \lambda$, and use the comparison measure W in both projection and transform. Its kernels are exactly K_N^0 , and χ_N becomes point evaluation at $v(t)$. Apart from the fixed positive factor $J(b/2)$, the residual transform is $J(iy)Q(v(iy))^N$ times

$$K_N^0(v, 1) - K_N^0(v, u_0) \frac{K_N^0(v(t), 1)}{K_N^0(v(t), u_0)}, \quad v = v(iy).$$

After division by N , Lemma 5.2 gives a limit uniform on the imaginary diameter. At $v = \pm i$, its sign is the sign of

$$\frac{1}{2} - \frac{1 - v(t)^2 u_0^2}{(1 + u_0^2)(1 - v(t)^2)},$$

which is strictly negative, uniformly for $0 \leq t \leq \lambda$. Indeed the second term is at least $1/(1 + u_0^2) > 1/2$. The remaining factors O are positive on both diameters, with uniform positive bounds on the compact sets in question.

Fourier inversion at zero integrates the bracket divided by N against

$$cN J(iy)Q(v(iy))^N dy, \quad c > 0.$$

This positive measure has mass bounded above and below and concentrates at $y \rightarrow \pm\infty$: on the imaginary diameter $Q \geq 0$, $1 - Q \asymp e^{-\pi|y|/b}$, and $J \asymp e^{-\pi|y|/b}$. The boundedness supplied by Lemma 5.2 and the negative endpoint limit therefore give the desired lower bound for W , uniformly in t .

To pass to μ , its exact density ratio to W is

$$\frac{d\mu}{dW} = \frac{(2u - 1)^2(2u + 1)}{4u(2u^2 - 1)}, \quad u = \cosh(\lambda\sqrt{x}).$$

It is positive and smooth on $[0, \infty)$ and eventually increasing. For $r(x) = (x + 2)/(x + 1)$, choose a fixed integer $j > 0$ so that $\mu/(r^j W)$ is increasing: on a compact interval the negative part of the logarithmic derivative is bounded, while $-(\log r)'$ is strictly positive; outside it the original derivative is already nonnegative. The ordered association input then bounds the expectation of $-e_0 c_3$ for μ below by that for $r^j W$.

Each multiplication of the base weight by r changes this expectation by only a bounded factor. Indeed it multiplies it by the ratio of the characteristic-product expectations of $\prod_i r(x_i)$ under $(A_N, G_{N,t})$ and $(F_{0,N}, G_{N,t})$ at the current base weight. The interlacing input compares the first to (U_N, U_N) and the second to (M_N, M_N) ; symmetric codimension-one compression compares these last two. The bounded range of $\log r$ gives a uniform constant at each step, independent of N, t . After the fixed number j of steps, the lower bound therefore remains positive. Finally integrate the positive $G_{N,t}$ mixture to obtain it for G_N . This proves the last line of (5.10).

The quotient. We can now finish Proposition 5.1. The denominator $r_N(c_3 - h_1)(0)$ in (5.2) is bounded away from zero for all sufficiently large N , because $-r_N c_3(0) \geq c$ and $|r_N h_1(0)| \leq N^{-1+o(1)}$. Its reciprocal has ℓ th differences $N^{-\ell+o(1)}$, by the discrete inverse rule and (5.10). In the numerator, $r_N h_0(-4)$ has differences $N^{-1+2b/\pi-\ell+o(1)}$, while the product of $r_N(c_3 - h_1)(-4)$ and $r_N h_0(0)$ has exactly the same bound. Apply the discrete product and quotient rules. Since $2b/\pi = 3/4$, the resulting exponent is $-1/4 - \ell + o(1)$, as asserted. For each fixed ℓ and exponent slack, the finitely many remaining values of N are absorbed in $C_{\ell,\eta}$.

6 Exact height and displacement atoms

We now turn strip regularity into estimates for a bridge with a prescribed terminal port. Throughout this section, p is the probability law of one irreducible bridge from a fixed port. Write H, Z, L , and D for its positive integral height, transverse displacement, number of visited triangle centers, and diameter. The vector $X = (H, Z)$ takes values on the actual row-staggered lattice; an atom outside this lattice is zero. For independent copies, put

$$S_k = X_1 + \cdots + X_k, \quad T_k = L_1 + \cdots + L_k.$$

Independence is used between whole pieces, never between coordinates of one piece.

We use the renewal identity and one-piece estimates stated earlier:

$$\sum_{h \geq 0} B_h z^h = \frac{1}{1 - \mathbb{E}_p z^H}, \quad B_0 = 1, \quad B_h \asymp (1 + h)^{-1/4}, \quad (6.1)$$

and, for $x, t \geq 1$ and $u \downarrow 0$,

$$\begin{aligned} p(H > x) + p(D > x) &\leq Cx^{-3/4}, \\ p(L > t) &\leq t^{-9/16+o(1)}, \quad 1 - \mathbb{E}_p e^{-uL} = u^{9/16+o(1)}. \end{aligned} \quad (6.2)$$

The first identity implies $1 - \mathbb{E}_p e^{-sH} \asymp s^{3/4}$. These are statements under p , before conditioning on an endpoint.

6.1 An atom of the one-piece height

The extra information supplied by strip regularity is one full power of height decay beyond the tail bound. This is the estimate that rules out bridges made from too few irreducibles at one prescribed endpoint.

Proposition 6.1. *For every $\eta > 0$ there is $C_\eta < \infty$ such that*

$$p(H = n) \leq C_\eta n^{-7/4+\eta} \quad (n \geq 1).$$

Proof. Let $b(z) = \sum_{m \geq 0} B_m z^m$. Fix $n \geq 2$, set $r = e^{-1/n}$, and write $s = \min\{1, n^{-1} + |\theta|\}$ for $-\pi \leq \theta \leq \pi$. The height tail gives

$$|1 - \mathbb{E}_p(re^{i\theta})^H| \leq C\mathbb{E}_p \min\{1, sH\} \leq Cs^{3/4}.$$

Thus $|b(re^{i\theta})| \geq cs^{-3/4}$. In particular b has no zero on this circle, as also follows from its renewal representation.

For every fixed integer $k \geq 0$ and every $\delta > 0$, the strip differences of Proposition 5.1 imply

$$\left| \partial_\theta^k b(re^{i\theta}) \right| \leq C_{k,\delta} s^{-3/4-k-\delta}. \quad (6.3)$$

Here are the details of the summation estimate. Choose a smooth dyadic partition of unity on the positive integers. On $m \asymp M$, discrete product rules and the strip differences bound every j th difference of $m^k B_m r^m$ times the cutoff by $C_{j,k,\delta} M^{k-1/4-j+\delta} e^{-M/(Cn)}$. Summation by parts j times, together with the trivial sum bound, gives

$$C_{j,k,\delta} M^{3/4+k+\delta} e^{-M/(Cn)} (1 + M|\theta|)^{-j}.$$

Choose $j > 3/4 + k + \delta$ and sum over dyadic M . The terms with $M \leq s^{-1}$ form a geometric sum; the remaining terms are bounded by the oscillatory factor or the exponential cutoff. This proves (6.3).

Differentiating the reciprocal twice now gives

$$\left| \partial_\theta^2 b(re^{i\theta})^{-1} \right| \leq C_\delta s^{-5/4-2\delta}.$$

Consequently its integral over $[-\pi, \pi]$ is at most $C_\delta n^{1/4+2\delta}$. Since $\mathbb{E}_p z^H = 1 - b(z)^{-1}$, the n th Fourier coefficient on this circle, integrated by parts twice, satisfies

$$r^n p(H = n) \leq \frac{1}{2\pi n^2} \int_{-\pi}^{\pi} \left| \partial_\theta^2 b(re^{i\theta})^{-1} \right| d\theta \leq C_\delta n^{-7/4+2\delta}.$$

The boundary terms vanish by periodicity, $r^n = e^{-1}$, and δ is arbitrary. $\square$

6.2 Two forms of displacement smoothing

We need both a planar atom bound and a bound summed over possible heights. The latter is stronger than merely summing the planar bound over all heights, since the height distribution has an infinite tail.

Proposition 6.2. *There is $C < \infty$ such that, for every integer $k \geq 1$,*

$$\sup_w \Pr(S_k = w) \leq Ck^{-8/3}, \quad \sum_{m \in \mathbb{Z}} \sup_{H(w)=m} \Pr(S_k = w) \leq Ck^{-4/3}. \quad (6.4)$$

Proof. We first establish a directional window for one piece. There are constants $0 < c_0 < C_0 < \infty$ and $c_1 > 0$ such that, uniformly over unit vectors e and all sufficiently large x ,

$$p\{c_0 x \leq |e \cdot X| \leq C_0 x\} \geq c_1 x^{-3/4}. \quad (6.5)$$

Use bridges of heights in $[x, 2x]$ and terminal ports in a transverse interval of length comparable to x . The point-to-point connector lower bound is $cx^{-5/4}$ at each such port. The two intervals of bounded positive and negative slope may be chosen so that their displacements have projection at

least a fixed multiple of x in any prescribed direction. Compactness of the unit circle makes the constants uniform. Summing over heights and ports gives critical mass at least $cx^{3/4}$.

By the renewal factorization, this is a sum of probabilities for S_k . The terms $k > Ax^{3/4}$ have total mass at most

$$\sum_{k > Ax^{3/4}} e^2 (\mathbb{E}_p e^{-H/x})^k \leq Cx^{3/4} e^{-cA}.$$

Fix A large. Some $k \leq Ax^{3/4}$ therefore has probability bounded below of producing an absolute projection comparable to x . The probability of an increment with $D > C_0 x$ is at most $CAC_0^{-3/4}$, while

$$\mathbb{E} \sum_{i \leq k} |e \cdot X_i| \mathbf{1}_{\{|e \cdot X_i| < c_0 x\}} \leq CAc_0^{1/4} x.$$

The second estimate follows by integrating the diameter tail; fixed lattice conversion constants are absorbed in C . Choose C_0 large and c_0 small. With a fixed positive probability an increment then lies in the window in (6.5). A union bound and $k \leq Ax^{3/4}$ prove that display.

Let $\varphi(\theta) = \mathbb{E} e^{i\theta \cdot X}$, using any fixed integer coordinates for the displacement lattice. An independent copy X' lies in a fixed bounded set with probability bounded below. Apply (6.5) in direction $\theta/|\theta|$, with x a sufficiently small fixed multiple of $|\theta|^{-1}$. On the resulting event, $|\theta \cdot (X - X')|$ stays between two positive constants less than π . Hence

$$1 - |\varphi(\theta)|^2 = \mathbb{E}[1 - \cos(\theta \cdot (X - X'))] \geq c|\theta|^{3/4}$$

near zero. The directional windows also show that the affine span of the support is two-dimensional. Its difference group is therefore a full-rank lattice subgroup. There are only finitely many points of the lattice torus where $|\varphi| = 1$; near each, its modulus is a translate of the modulus near zero. On the remaining compact set it is strictly less than one. Fourier inversion gives

$$\sup_w \Pr(S_k = w) \leq C \int_0^1 e^{-ckr^{3/4}} r dr + Ce^{-ck} \leq Ck^{-8/3}.$$

This also explains why no aperiodicity assumption beyond the actual support lattice is needed.

It remains to sum the heightwise supremums. For a nonnegative measure q on the displacement lattice, set

$$\|q\|_{1,\infty} = \sum_m \sup_{H(w)=m} q(w).$$

Convolution satisfies $\|q * r\|_{1,\infty} \leq \|q\|_{1,\infty} \|r\|_1$. The first bound of (6.4) immediately bounds the sum over $m \leq 2k^{4/3}$ by $Ck^{-4/3}$.

For a dyadic shell $m \asymp \Lambda k^{4/3}$, $\Lambda \geq 2$, split the k pieces into sixteen independent blocks with sizes comparable to k ; bounded k can be absorbed into the constant. Each block has planar atoms at most $Ck^{-8/3}$. Also a block height exceeds $t \geq k^{4/3}$ with probability at most $Ckt^{-3/4}$: split at increments larger than t and use the truncated first moment for the rest.

Suppose first that one block has height at most $k^{4/3}\sqrt{\Lambda}$. For large Λ , another has height at least $c\Lambda k^{4/3}$. The first restricted block has $\|\cdot\|_{1,\infty}$ at most $C\sqrt{\Lambda} k^{-4/3}$ by its planar atom bound, while the latter event has probability at most $C\Lambda^{-3/4}$. Convolve with the remaining blocks and sum over the finitely many choices. This gives $C\Lambda^{-1/4} k^{-4/3}$ for the shell. If all sixteen block heights exceed $k^{4/3}\sqrt{\Lambda}$, impose the planar atom on one block and the probability bound $C\Lambda^{-3/8}$ on each of the other fifteen. There are $O(\Lambda k^{4/3})$ heights in the shell, so this case costs at most $C\Lambda^{-37/8} k^{-4/3}$. Both bounds are summable over dyadic Λ . Increasing the constant handles the finitely many small Λ and bounded k , completing the proof. $\square$

7 Prescribed ports on parallel sides

The strip estimates above concern a fixed initial port and all terminal ports. We now prescribe the second port as well. The connector construction gives the required normalization: when the height is h and the transverse displacement is $O(h)$, the critical mass between the two ports is at least $ch^{-5/4}$. The constant is uniform over fixed compact nondegenerate scaled geometries. The same construction lies in a convex pure-sided domain if both ports have clearance comparable to h from every side other than the opposite parallel sides containing them. We will show that the exceptional mass in the entire strip is $o(h^{-5/4})$.

Theorem 7.1. *Fix a bound on the ratio of transverse displacement to height. For compatible prescribed ports on two parallel pure cuts at integral height $h \rightarrow \infty$, normalize the critical bridge weights by their total mass. For every $\epsilon > 0$,*

$$\Pr\{h^{4/3-\epsilon} \leq L \leq h^{4/3+\epsilon}\} \rightarrow 1.$$

The convergence is uniform under the fixed displacement bound. It also holds in convex pure-sided domains in a fixed compact family of nondegenerate scaled geometries, with the ports on opposite parallel sides and with order- h clearance from every other side.

Proof. It is enough to treat $0 < \epsilon < 1$. Let w be the prescribed displacement, with $H(w) = h$. Unique factorization writes its bridge mass as $\sum_{k \geq 1} \Pr(S_k = w)$. We first discard very small and very large numbers of pieces without imposing a length restriction.

Fix a small $\xi > 0$. Put $K = h^{3/4-\xi}$ and $a = h^{1-\tau}$, where $0 < \tau < 8\xi/21$. For $k \leq K$, the event $\sum_{i \leq k} H_i = h$ with every $H_i < a$ has probability at most

$$e^{-h/a} (\mathbb{E}[e^{H/a}; H < a])^k \leq \exp\{-h^\tau + Cka^{-3/4}\}.$$

We used $e^t - 1 \leq et$ for $0 \leq t \leq 1$ and $\mathbb{E}[H; H < a] \leq Ca^{1/4}$. Since $ka^{-3/4} \leq h^{-\xi+3\tau/4}$, summing these probabilities over $k \leq K$ is smaller than every inverse power of h .

If some $H_i \geq a$, fix its index and condition on the other pieces. Proposition 6.1, followed by the heightwise bound of Proposition 6.2, gives

$$\begin{aligned} & \Pr\{S_k = w, H_i \geq a\} \\ & \leq \sup_{n \geq a} p(H = n) \sum_{m} \sup_{H(v)=m} \Pr(S_{k-1} = v) \leq a^{-7/4+o(1)} C(1+k)^{-4/3}. \end{aligned}$$

For $k = 1$ the remaining sum is interpreted as the unit mass at zero. Summing over the candidate index and then over $k \leq K$ gives

$$Ca^{-7/4+o(1)} K^{2/3} = h^{-5/4-2\xi/3+7\tau/4+o(1)} = o(h^{-5/4}).$$

This is the step that needs both the one-piece height atom and the heightwise supremum estimate.

For $k > h^{3/4+\xi}$, height discount alone gives

$$\sum_{k > h^{3/4+\xi}} \Pr\left\{\sum_{i \leq k} H_i = h\right\} \leq e \sum_{k > h^{3/4+\xi}} (\mathbb{E}e^{-H/h})^k \leq Ch^{3/4} e^{-ch^\xi}.$$

We are left with $h^{3/4-\xi} < k \leq h^{3/4+\xi}$.

Take ξ small relative to ϵ . The length deficit in (6.2), with $u = h^{-4/3+\epsilon}$, shows uniformly over these k that

$$\Pr\{T_k < h^{4/3-\epsilon}\} \leq e (\mathbb{E}e^{-uL})^k \leq \exp\{-h^{9\epsilon/16-\xi-o(1)}\}.$$

Summing in k is negligible.

For the upper tail, split the pieces into two independent batches with sizes comparable to k . If $T_k > h^{4/3+\epsilon}$, one batch has length exceeding $t = \frac{1}{2}h^{4/3+\epsilon}$. The length tail and truncation yield, for any fixed small $\eta > 0$,

$$\Pr\{\text{that batch has length} > t\} \leq C_\eta k t^{-9/16+\eta} \leq h^{-\delta}$$

for some $\delta > 0$, provided ξ, η are sufficiently small relative to ϵ . On the other, independent batch impose the planar atom at the displacement needed to reach w . This costs at most $Ck^{-8/3}$. Thus the sum of long-path masses over the remaining k is bounded by

$$Ch^{-\delta} \sum_{k > h^{3/4-\xi}} k^{-8/3} \leq Ch^{-5/4+5\xi/3-\delta} = o(h^{-5/4}),$$

after decreasing ξ once more. This use of separate batches avoids any independence assertion within a piece or after endpoint conditioning.

Both length exceptions therefore have absolute strip mass $o(h^{-5/4})$. Divide by the connector lower bound. Restricting to a finite domain can only decrease the exceptional mass, and its own connector lower bound has the same order. This proves both assertions. $\square$

8 Prescribed ports on nonparallel cuts

The difference from the preceding two geometries is that a normal bridge prefix changes the distance to the opposite support. Summing over that support first gives an almost exhaustive prefix construction. We must then show that its small exceptional mass stays small at a specified port. All estimates below are uniform on a fixed compact set of the nondegenerate wedge geometries in Theorem 1.1. We continue to use $d = \sqrt{3}/2$ and the winding spin $\sigma = 3/8$.

Theorem 8.1 (Tilted prescribed endpoints). *In a pure wedge of angle $\pi/3$ or $2\pi/3$, take one prescribed port on each boundary ray. Suppose they have order- R clearance to the opposite cut and distance $O(R)$ from the apex. Under the critical chord law their length is $R^{4/3+o(1)}$ in probability. The conclusion persists after restriction to the hexagons of Theorem 1.1 with the same side-interior limiting data.*

We now treat the case when the supporting sides at the two marked midpoints have distinct, nonparallel pure directions. Write W for the full wedge given by the two corresponding row cuts, and n, m for their inward unit normals in source and target order. Write a, Q for the midpoints. Both endpoints have clearance to the other cut of order R , and their distances from the apex are $O(R)$. We let $R \rightarrow \infty$; constants may use fixed compact bounds on this nondegenerate scaled geometry. Until restricting the domain at the end we count paths in W . Denote their mass at a specific endpoint U on the target ray by $K(a, U; W)$ (abbreviated to K_U).

First we explain exhaustion by normal bridge prefixes when *summing* over the target ray. Put r equal to the distance from a to the target support line in row-spacing units, and $F(r)$ equal to $\sum_U K_U$ along that ray. With directions fixed, the parameter alone determines this mass, by lattice translation (fix the start and its entire line); its possible values here are positive half-odd levels. This function decreases: keeping the source line fixed, the wedge expands as r increases, and the real contour identity gives

$$1 = \cos(\sigma\pi)A(r) + \cos(\sigma\theta)F(r)$$

where A sums exits back to the initial support, and θ is the opening angle ($\pi/3$ or $2\pi/3$). Indeed turns to the other side have magnitude θ ; distant truncating exits vanish by the diameter bound. Thus monotonicity follows from that of A . Also $F(r) \lesssim (1+r)^{-1/4}$.

Use the ordinary iid irreducible pieces directed along n . For each deterministic k , call the prefix admissibly small when the sum of the k diameters J (enlarged by a constant if needed) is less than a sufficiently small fixed multiple of the source clearance to the other line. Write w for its displacement. Then

$$F(r) \geq \mathbb{E}[\mathbf{1}_{\text{small}} F(r + m \cdot w/d)],$$

by using the smaller wedge W' with initial support now passing through $a' = a + w$, appending its chords at a' . All points along the prefix avoid the target line. Weights multiply, and one recognizes the prefix by the first k positive single-crossing levels parallel to the initial support (the last one also separates the appended path strictly).

We record why $F(r + o(r)) - F(r) = o(r^{-1/4})$. For any large scale q there is $k \leq Cq^{3/4}$ with probability bounded below of $m \cdot w/d \leq -b_0q$ for some fixed $b_0 > 0$. Indeed the source bridges of heights in $[q, 2q]$ have summed mass $\gtrsim q^{3/4}$ with such displacement, by Lemma 3.4 with second direction $-m$. Equivalently use point-to-point corridors to macroscopic intervals on the top in a bounded range of slopes with that strict projection. Terms with more than $Cq^{3/4}$ pieces cost at most $O(q^{3/4}e^{-cC})$ by the height Laplace deficit. This proves the assertion via renewal. At these orders k , probability that the diameter sum exceeds Tq is $O(T^{-3/4})$ for $T \geq 1$, by the tail bound and truncation.

If there is a drop at least $\epsilon r^{-1/4}$ between $r-l, r+l$ with $1 \leq l = o(r)$ along a sequence (round levels throughout), choose a large fixed A_0 , then $l \ll q \ll r$ with variation on $[r + b_0q/4, r + (A_0 + b_0 + 1)q]$ equal to $o(r^{-1/4})$. Indeed one can test arbitrarily many disjoint such geometric bands and use monotonicity. Apply the inequality at j near $r + b_0q/2$ with the indicated k . On small prefixes with $|m \cdot w/d| \leq A_0q$ the joining factor is at least $F(j) - o(r^{-1/4})$; if the shift is $\leq -b_0q$ it gains at least $\epsilon r^{-1/4}$. Smallness has probability tending to one; the remaining offset loss is arbitrarily small in probability by choice of A_0 . Since $F(j) \lesssim r^{-1/4}$, this is impossible.

In particular take $k = \lfloor R^{3/4-\eta} \rfloor$, fixed small $\eta > 0$. In the given geometries use a common smallness cutoff on $\sum J$ at a sufficiently small fixed multiple of R (less than a small fraction of both endpoint clearances). Define exceptional paths to be all chords not so constructed with this prefix and a chord in W' , and denote their mass at U by E_U . By the last continuity, truncation and $r \asymp R$, the constructed total in expectation over small prefixes differs from $F(r)$ by $o(R^{-1/4})$. Hence $\sum_U E_U = o(R^{-1/4})$.

8.1 Visits near the prescribed target

The prefix construction has so far controlled only the sum of exceptional masses. To pass to a single target, we need a bound for paths that approach that target before ending a specified distance away. We give a local visit estimate. For $q \leq cR$, consider paths from a to points P on the target ray at distance between q and $2q$ from Q , also coming within t of Q , $1 \leq t \leq q$. Their total mass satisfies

$$H_q(t) \lesssim R^{-1/4}(t/q)^{3/8}. \quad (8.1)$$

Here and below $c > 0$ can be taken small relative to the compact geometry bounds. Use the fixed-source construction of Proposition 3.6, with the side-1 outer landing exactly at a and side-2 outer landing free. Take translation scale a small fixed multiple of q , so all bridge portions and translations have excursion less than (say) $q/20$, with gaps between original and translated levels at each end of order q . If q is bounded the estimate already follows from diameter. The first summed

indicator mass is at least $c_1 q^{1/2} H_q(t)$, using $\gtrsim q^{3/4}$ summed bridge mass from a to its translates and $\gtrsim q^{-1/4}$ on side 2. The target measure (paths from a to the other ray) has mass $O(R^{-1/4})$.

For simultaneous translations v, v' , split at both support crossings on each branch in level order as in the chord construction. The orders may differ at the two ends. All initial crossings still precede both terminal crossings, by localization of the two exterior branches. Write $P_v, P_{v'}$ for the terminal ports of the two input chords; each lies in the bin at distance between q and $2q$ from Q . For $t \leq q/10$, a visit near either translated landmark $Q + v$ or $Q + v'$ cannot lie on an exterior branch. The initial branches remain at distance of order R from these landmarks. Each terminal branch has excursion less than $q/20$ from its endpoint $P_v + v$ or $P_{v'} + v'$; the endpoint bin and the small translation scale keep every such endpoint at distance comparable to q from both landmarks. Thus both visits occur after both initial crossings and before both terminal crossings.

This common central portion stays above both target cuts. Its visit within t of the landmark on the lower cut therefore forces $|m \cdot (v - v')| \leq Ct$. Moreover, the chord between the innermost cuts retains the visit to the landmark belonging to its own terminal cut. Its terminal point is the translated P from that same list, so its distance to the landmark is still in the prescribed bin; its initial point is also prescribed. The endpoints and landmark have only shifted by $O(q)$, so this mixed-cut wedge has comparable compact geometry bounds. If the visit estimate is known with exponent α for these geometries, its central mass is consequently at most $CR^{-1/4}(t/q)^\alpha$. The diameter bound starts this argument at $\alpha = 0$.

The mixed difference factor now sums to $O(t^{1/2})$:

$$\sum_{z: |m \cdot z| \leq Ct} D_n(z)(1 + |m \cdot z|)^{-1/4} \leq C' t^{1/2},$$

with D_n defined before Lemma 3.5. The projected concentration and negative-quarter-moment bounds in (3.4) bound the contribution from j pieces by $C \min(j^{-1/3}, (1+t)^{3/4} j^{-4/3})$. Summing in j by renewal, including ties, proves the restricted estimate. The outer side-2 bridge costs $O(q^{-1/4})$; the side-1 outer cost $D_n(v_{\text{out}})$ sums to $O(q^{3/4})$, where v_{out} gives the side-1 outer crossing. Drop mutual avoidance constraints between the difference, center and outer portions and multiply their upper bounds. The summed second moment is at most

$$C q^{1/2} t^{1/2} R^{-1/4} (t/q)^\alpha.$$

Cauchy–Schwarz proves the assertion with exponent $1/4 + \alpha/2$ (the remaining $t \geq q/10$ always follows by diameter). Applying twice proves the displayed estimate; the shifts of supports in this finite bootstrap are $O(q)$ each, so one can enlarge the compact bounds a little and decrease the allowed relative scale c at each step.

8.2 An endpointwise comparison

We adapt the exterior-arc comparison from [16, proof of Proposition 7.1] to the nonparallel supports. Take two individual endpoints P, Q , with P nearer the apex and $|P - Q| \in [q, 2q]$. Add a formal connection from P to Q in the exterior of the target half-plane: a semicircle tangent to the normal rays at its anchors. This arc is not a honeycomb walk. Assign it modulus weight one, no visited-center length, and the phase $e^{i\sigma\Delta W}$ of its geometric turn. Sum only paths using the arc exactly once; all their other edges and vertex weights are ordinary. The arc's exterior lies outside W , so it meets no ordinary path away from its anchors. Figure 2 records its orientation. Arrange the coordinates so the initial ray points right, and the target ray lies counterclockwise at angle θ . Thus

$a \rightarrow P, a \rightarrow Q$ turn by θ ; the arc $P \rightarrow Q$ turns by $-\pi$. For a path $\omega : a \rightarrow Q$ define

$$f_W(P; \omega) = \frac{1}{2 \cos(\sigma\pi)} - \sum_{b' \in (P, Q)} \sum_{\substack{\zeta: P \rightarrow b' \text{ in } W \\ \zeta \text{ disjoint from } \omega}} \kappa^{L(\zeta)}.$$

For paths to P use $f_W(Q; \omega)$ analogously with continuation from Q and the same open interval. Then (integrals denoting critical weighted sums)

$$\left| \int_{a \rightarrow Q} f_W(P; \omega) - \int_{a \rightarrow P} f_W(Q; \omega) \right| \leq CR^{-1/4}(K_Q + K_P).$$

Indeed the flux delivered by arc traversals before further continuation equals the outgoing flux using the arc once. Prefix cancellations at all ordinary vertices preserve the count of arc uses under unfinished-loop reversal; the whole arc is well after the initial cut so the source cannot be inside a loop. Use first a large convex truncation; distant contributions vanish by ordinary diameter bounds. After rotation by $e^{-i\sigma\theta}$ the imaginary input is $\sin(\sigma\pi)(K_Q - K_P)$. Outputs back to the initial support have modulus cost at most $CR^{-1/4}(K_Q + K_P)$. Outputs on the target support from an initial traversal $a \rightarrow Q$, then the arc to P , must end nearer than Q by noncrossing. If ending even before P their relative phase after rotation is 1, while in the open interval it has imaginary part $\sin(2\sigma\pi)$ (final arch positive turn π). Similarly from $a \rightarrow P$ the only target-side terms with imaginary part end in (P, Q) , with opposite sign. Neither anchor can be such a final exit (used triangles). Dividing by $\sin(2\sigma\pi)$ proves the comparison.

Reflection and the half-plane contour identity give total arch mass $1/(2 \cos(\sigma\pi))$ in each direction. Together with the half-plane arch estimates, this gives, for paths to Q ,

$$c'q^{-1/4} \leq f_W(P; \omega) \leq C(1 + \min(q, \text{dist}(P, \omega)))^{-1/4} \quad (8.2)$$

and analogously after exchanging the endpoints. Indeed all completions counted there are among half-plane arches in one direction, but omitting at least the long-offset tail of mass $\gtrsim q^{-1/4}$. All sufficiently small diameter ones up to scale the indicated minimum are included, by clearance and disjointness, and the diameter tail bounds the loss.

With q a sufficiently small fixed fraction of R , the comparison absorbs the $O(R^{-1/4})K_Q$ term. On the other side average over the order q choices of P . The visit bound $H_q(t)$, with exponent $> 1/4$, pays for the upper estimate on $f_W(Q; \omega)$ by dyadic summation. More explicitly the total over P of these upper integrals costs $O(R^{-1/4}q^{-1/4})$, including unit and smaller distances and also paths always at distance $> q$. This yields the uniform bound $K_Q \lesssim R^{-5/4}$.

8.3 Removing the exceptional mass

The bound for each individual kernel is now available. Apply the same comparison to the exceptional set left by the prefix construction. The same absolute comparison holds with the two left integrals restricted to exceptional paths and the right error still $CR^{-1/4}(K_Q + K_P)$, with C independent of $q \leq cR$. To see this, subtract the comparison (in signed-difference form with its error) in W' , starting at a' , integrated over small prefixes. Its right error costs at most the same amount by the prefix injection. And the difference of the two f factors for a central path and full constructed path costs only $O(R^{-1/4})$ each, since central completions in the sum stay beyond the prefix cut and avoid the prefix too, and any discarded full-domain completions must reach that cut from clearance of order R . The constants are uniform by smallness of the prefixes.

$$P \rightarrow Q : \quad \Delta W = -\pi, \quad \text{modulus } 1$$

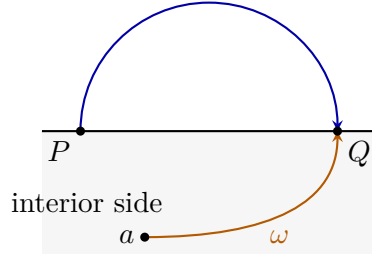


Figure 2: Local orientation at the target support, rotated to horizontal. The blue semicircle is a formal exterior connection, not an added lattice path. The orange path arrives at Q from the interior; its continuation through the arc uses the reverse orientation. The schematic suppresses the source support and the lattice.

Fix q of order ρR for small fixed $\rho > 0$, and put

$$e_R = R^{1/4} \sum_U E_U \longrightarrow 0.$$

Let $I_R(q)$ be the sum over the order- q choices of P of $\int_{a \rightarrow P, \text{exceptional}} f_W(Q; \omega)$. To control its possibly singular weight, fix $0 < \delta < 1/2$. Paths with $\text{dist}(Q, \omega) \geq \delta q$ contribute at most $C(\delta q)^{-1/4} \sum_P E_P$, by (8.2). For the remaining paths, discard the exceptional restriction and sum (8.1) over dyadic distance bins below δq . The exponent $3/8 - 1/4 = 1/8$ is positive, so, including the final bounded-distance bin, for R sufficiently large at fixed ρ, δ this gives

$$I_R(q) \leq CR^{-1/4} q^{-1/4} (\delta^{-1/4} e_R + \delta^{1/8}).$$

First let $R \rightarrow \infty$ at fixed ρ, δ , then let $\delta \downarrow 0$. Thus $I_R(q) = o(R^{-1/4} q^{-1/4})$ for each fixed ρ . Averaging the exceptional comparison over its order- q ports, using the lower bound in (8.2) and the individual kernel upper bound, now yields

$$q^{-1/4} E_Q \leq o(R^{-1/4} q^{-5/4}) + CR^{-1/4} R^{-5/4}.$$

Here the little-oh may depend on ρ , but the comparison error constant C does not. In particular

$$\limsup_{R \rightarrow \infty} R^{5/4} E_Q \leq C\rho^{1/4}.$$

Letting $\rho \downarrow 0$ proves $E_Q = o(R^{-5/4})$.

8.4 The prescribed length law

Finally count lengths violating either power bound $R^{4/3 \pm \epsilon}$. Up to mass $o(R^{-5/4})$, the prefix construction applies at the prescribed Q . Under the iid law its prefix length has exponent

$$\frac{3/4 - \eta}{9/16} = \frac{4}{3} - \frac{16\eta}{9}$$

in probability. On small prefixes the center factor is at most $CR^{-5/4}$, uniformly in their displacements. Therefore an iid event of probability $o(1)$ contributes constructed mass $o(R^{-5/4})$, without asserting

independence after fixing Q . Choose $\eta < 9\epsilon/16$. This proves the short-length bound and also makes prefixes longer than $\frac{1}{2}R^{4/3+\epsilon}$ negligible.

For the appended chord, any specific prefix displacement costs $O(k^{-8/3})$ by Proposition 6.2. Align all new wedges by lattice translations τ_w ; their apex intersections are triangular lattice vertices. In the common wedge the endpoints are

$$x_w = a + w + \tau_w, \quad y_w = Q + \tau_w, \quad w = (x_w - y_w) - (a - Q).$$

Thus each endpoint pair determines its prefix displacement. Since the angles are fixed and $|w| = O(R)$ for small prefixes, the apex shift τ_w is also $O(R)$. In particular there are only $O(R)$ possible aligned source ports, all at distance $O(R)$ from the apex.

We may now sum freely over the endpoint pairs. For appended chords of diameter above $S_* = R^{1+\lambda}$, the one-source diameter bound and the number of sources give $CRS_*^{-1/4}$. All smaller-diameter chords lie in a common convex pure truncation of diameter $O(S_*)$. The all-endpoint length-mass bound there shows that those of length above $\frac{1}{2}R^{4/3+\epsilon}$ have total mass at most $CR^{-4/3-\epsilon}S_*^{25/12+o(1)}$. Multiplying these two estimates by $k^{-8/3}$, respectively, gives

$$CR^{-5/4-\lambda/4+8\eta/3}, \quad R^{-5/4-\epsilon+25\lambda/12+8\eta/3+o(1)}.$$

Choose first $\lambda > 0$ so that $25\lambda/12 < \epsilon$, then $\eta > 0$ sufficiently small that both exponents are strictly below $-5/4$, while retaining $\eta < 9\epsilon/16$. Both appended-chord exceptions are consequently $o(R^{-5/4})$.

Restriction to a hexagon in Theorem 1.1 can only decrease these absolute exceptional masses. Its own normalizer is at least $cR^{-5/4}$, by Proposition 3.1 and the endpoint clearances. Dividing proves that both tails vanish in probability in the restricted domain as well as in the full wedge.

This proves Theorem 8.1. Together with Theorem 4.1 and Theorem 7.1, it proves Theorem 1.1: the two side supports are identical, parallel, or nonparallel, and the connector lower bound supplies the normalizer in each finite hexagon.

9 Mean length of a prescribed half-plane arch

The same-side probability theorem does not control mean length. We give a separate marked-arm proof, whose length-weighted contributions are summable over arch heights at each prescribed gap. It uses the calibrated finite method and its renewal consequences.

Theorem 9.1 (Every-gap half-plane mean length). *For two half-plane boundary ports separated by each integer $g \rightarrow \infty$, the probability law $\kappa^{L(\gamma)}/K_g$ satisfies*

$$L(\gamma) = g^{4/3+o(1)} \quad \text{in probability,} \quad \mathbb{E}L(\gamma) = g^{4/3+o(1)}.$$

The finite calibrated estimates in [12, Theorem 1.2 and Proposition 11.1] give $B_h \asymp h^{-1/4}$, $K_g \asymp g^{-5/4}$ with its lower bound retained at diameter at most Ag for fixed A , an exponential lateral cutoff for a fixed-height strip, and total chord length mass at most $CR^{25/12}$ in a convex box of size R . Their precise confined length window is not replaced by a pointwise endpoint estimate here. The probability inputs below are derived from this finite package in [15, Proposition 4.1, Lemma 4.2, Lemmas 7.2 and 7.4, Section 7.3, and Theorem 7.1]. Proposition 4.1 and Lemma 4.2 give the opening height, diameter, and length-deficit estimates; Lemma 7.2 gives the atom bounds; Lemma 7.4 and Section 7.3 give the censored survival test and its shared-turn application; Theorem 7.1 gives the absolute short-path bound.

Proposition 9.2 (Calibrated renewal inputs). *For the port-rooted irreducible law, write Y, Z, L, D for integral height, lateral displacement, length and diameter. Then $2Z$ is integral, $\Pr(D > r) \leq Cr^{-3/4}$, $1 - \mathbb{E}e^{-tY} \asymp t^{3/4}$, and $1 - \mathbb{E}e^{-tL} = t^{9/16+o(1)}$. In addition:*

- (i) *For k independent whole pieces, $\sup_j \Pr(S_Y(k) = j) \leq Ck^{-4/3}$ and $\sup_w \Pr(S_{(Y,Z)}(k) = w) \leq Ck^{-8/3}$. The off-axis bridge construction gives positive mass in directional windows; all atoms use the actual lattice.*
- (ii) *Start two independent sequences at the same bottom port through the same first center v . Retain completed pieces at heights strictly below s , excluding the first piece reaching or passing s . Fix left and right roles. At every pair of renewal levels $0 < d_1 < d_2 < s$ from the left and right histories respectively, require that the prefixes be disjoint after v , leave on opposite sides of v , and that every right-prefix crossing at d_1 lie strictly right of the left terminal coordinate. The probability of all these requirements is at most $C_\eta(1+s)^{-3/4+\eta}$ for each $\eta > 0$. At a shared turn this applies with $s = 1$ vacuously, or with s sufficiently below both adjacent arm heights that distant endpoint adjustments do not affect the tests. For the two arms used below, extending both to the same top port duplicates just their highest center. The resulting factor of κ is absorbed by an absolute constant; no shape-dependent trimming is used in this application.*
- (iii) *For every sufficiently small fixed $\xi > 0$, some $c, C > 0$ give the absolute critical mass estimate*

$$\sum_{\substack{\omega \text{ rooted at a fixed center} \\ R \leq \text{diam } \omega \leq 2R, L(\omega) \leq R^{4/3-\xi}}} \kappa^{L(\omega)} \leq Ce^{-R^c}.$$

9.1 The two arms and the dyadic target

Split an arch at a chosen highest vertex. Its two real incident edges go downwards. Extending both arms to the directly adjacent top wall port makes them strict downward bridges, duplicating just that vertex. Their critical product weight therefore differs from the arch weight by a fixed factor. We count the two shapes from a reference top port, impose equal heights and transverse displacement difference g , and translate the first bottom endpoint to its prescribed position. This entails no additional placement sum.

The two ends of this decomposition impose different constraints. At the top the arms share a turn, so Proposition 9.2(ii) applies. At the bottom they start at distinct ports a distance g apart; we will prove a separate weak-order estimate there. To sum length, mark one irreducible on one arm with its length, and reserve independent pieces away from both ends and the mark to match height and displacement. Adding the two choices of rewarded arm bounds the full length weight.

Put $d_* = 4/3$ and $a = 3/4$. For a dyadic scale H , let

$$\mathcal{M}_g(H) = \sum_{\substack{\gamma: 0 \rightarrow g \text{ a half-plane arch} \\ H \leq h(\gamma) < 2H}} L(\gamma) \kappa^{L(\gamma)},$$

where $h(\gamma)$ is the integral height of the adjacent top wall just described. The target estimate is

$$\mathcal{M}_g(H) \leq C_\eta g^{1/4} H^{d_* - 2a + O(\eta)} \quad (H \geq g / \log^2 g). \quad (9.1)$$

The smaller height scales will be negligible by strip confinement. Since $d_* - 2a = -1/6$, summing this bound for sufficiently small η gives total length mass $g^{1/12+o(1)}$. Division by $K_g \gtrsim g^{-5/4}$ then gives the required mean upper bound.

9.2 A local truncated reward

The first ingredient bounds the cost of the piece carrying the length weight.

Lemma 9.3 (Truncated length reward). *For the strict irreducible law, as $H \rightarrow \infty$,*

$$\mathbb{E}[L; Y \leq 2H] \leq H^{d_* - a + o(1)}.$$

Proof. Write $\mathcal{L}_h$ for the total length-weighted mass of strict bridges of height h from one fixed port, with terminal port summed. Insert a marked irreducible of height at most $2H$ between two arbitrary strict bridges of heights at most H . Unique irreducible factorization gives

$$\left(\sum_{0 \leq u \leq H} B_u \right)^2 \mathbb{E}[L; Y \leq 2H] \leq \sum_{0 \leq h \leq 4H} \mathcal{L}_h.$$

Indeed the same output can have several eligible marked pieces, but their lengths sum to at most its full length. The bridge sum on the left is of order H^a .

For height h , restrict first to diameter at most $h \log^2 h$. Translate the source to order h positions and apply the calibrated convex-box length bound in a box of size $O(h \log^2 h)$. This gives $\mathcal{L}_h \leq h^{13/12 + o(1)}$ for the restricted bridges. The larger diameter tail is negligible by the exponential strip cutoff and the area bound on length. Bounded strip heights also have exponential diameter tails: extend them a fixed amount to a sufficiently large fixed height. Consequently

$$\sum_{h \leq 4H} \mathcal{L}_h \leq H^{a + d_* + o(1)}.$$

The insertion inequality now proves the claim. $\square$

9.3 Survival of the order at separated bottom ports

The shared-turn estimate does not apply at two different starting ports. We therefore prove the separate weak-order estimate required at the bottom ends of the arch. This estimate uses the calibrated finite inputs throughout, independently of the off-axis construction in Section 4.

Lemma 9.4 (Weak order from separated ports). *Start two independent upward irreducible sequences at bottom ports a distance $g \geq 1$ apart, assigning left and right roles by their starts. Let $\Pi_g(S)$ be the probability that their horizontal endpoint order is preserved weakly at every common renewal height strictly below S . For $S \geq 2$,*

$$\Pi_g(S) \leq C(g/S)^{1/4} \sqrt{\log(2 + S)}.$$

Proof. Their strictly positive common renewal levels themselves form renewals; write (T, W) for one step, recording height and the signed change of separation. Pairs reaching the same height split there freely. Their renewal mass at j is $V_j = B_j^2 \asymp (1 + j)^{-1/2}$, so the first common return has total mass one and the pairs restart independently. Write $b = 1/2$. The transform gives $1 - \mathbb{E}e^{-T/r} \asymp r^{-b}$. Exchanging the two sequences preserves the joint law of (T, W) and replaces W by $-W$; hence height damping also preserves displacement symmetry. We first establish

$$\Pr(|W| > r) \leq Cr^{-b}, \quad \mathbb{E} \min(1, |W|/r) \gtrsim r^{-b}.$$

Pairs in a height bin around $c_0 r$, with c_0 small, carry summed mass $\gtrsim r^b$ within individual diameter $r/3$ by tightness, hence with all W increments bounded by r . Bounding this by the restricted

renewal sum proves the first claim. For the second take integral height $h \asymp r$. We need strict bridges of height h with endpoint displacement of a fixed nonzero sign and size at least ch , each of total mass $\gtrsim h^{-1/4}$. The half-plane continuation in [12, Section 11, Recoverable half-plane growth] uses two pure inward normals with dot product $1/2$. For each fixed sufficiently small transverse tolerance, it continues from initial scale r to the exact wall of the second normal at distance $S \geq Kr$, with its terminal port free. Its mass is at least $cS^{-1/4}(S/r)^{-C}$ and diameter at most $C'S$. All centers stay after the initial wall and before the terminal wall, the continuation has the prescribed relative transverse tolerance, and its joining cuts are recoverable. The constants and lower threshold for r may depend on that tolerance.

Apply it first with normals y, u at angle $\pi/3$, initial scale c_*h , and each exact u -wall whose level lies in $[(2 - 2\delta_*)h, (2 - \delta_*)h]$, always summing its terminal ports. For the later continuation with normals u, y , fix in advance a small transverse tolerance and a sufficiently small initial-scale fraction compatible with its target-to-initial-scale requirement. Its diameter constant is then fixed independently of δ_* . Choose δ_* small in terms of that constant, and then choose the first continuation's tolerance and c_* sufficiently smaller, also satisfying its required scale ratio.

The first continuation gains y -height with slope $1/2$, so it stays in $0 < y < h$, ends with remaining height between $\delta_*h/4$ and $2\delta_*h$, and has lateral displacement at least ch . Each target wall has mass $\gtrsim h^{-1/4}$ because all scale ratios are fixed. Extract the prefix at the lowest qualifying singly crossed u -cut in this bin, using these confinement and terminal-position tests. The mass summed over target levels is $\gtrsim h^{3/4}$, whereas all suffixes after a given prefix cost at most $\sum_{j \leq Ch} B_j \leq Ch^{3/4}$. Distinct prefixes therefore have mass $\gtrsim 1$.

From each prefix apply the u, y continuation with the preselected tolerance and initial-scale fraction to reach the wall $y = h$ exactly, again with terminal port free. It stays beyond the joining u -cut, has diameter $O(\delta_*h)$ and mass $\gtrsim h^{-1/4}$. By the choice of δ_* , it preserves the nonzero lateral displacement and avoids $y = 0$. The joining cut is recovered by its first-qualifying rule. This proves the asserted strict-bridge mass within the calibrated method.

Reflect the construction for the second bridge. The critical product mass with total transverse difference $\geq ch$ in absolute value is therefore $\gtrsim h^{-1/2} = h^{b-1}$. Bound using subadditivity of $\min(1, |\cdot|/r)$ over common steps and renewal insertions, costing at most $\max_{s \leq h} \sum_{i+j=s} V_i V_j$ (bounded) times the asserted expectation. This proves the claim. As in the local bounds before (by tail truncation and symmetrization) these imply $\Pr(cr \leq |W| \leq r) \gtrsim r^{-b}$ and $1 - |\phi_W(u)|^2 \gtrsim |u|^b$ near zero for the characteristic function. Thus independent sums of k such increments have atoms $\leq Ck^{-1/b}$, by lattice Fourier inversion (nonconstant support on a discrete grid).

Counting near-minimum times. The distinction between weak and strict survival for lattice increments is essential; compare [18, Section 2.2]. We use the following direct factorization, which counts all near-minimum times. Kill at an independent exponential height E of mean S . The transverse walk has symmetric subprobability step kernel

$$Q_S(w) = \mathbb{E}[e^{-T/S}; W = w], \quad q_0 = 1 - \sum_w Q_S(w) \asymp S^{-b}.$$

Let p be its probability of having all partial sums at least $-g$ before death. Write N_g for the number of indices, including time zero, lying within g of its global minimum. Then $\mathbb{E}N_g = p^2/q_0$. Here is a count that also handles ties. If $a_j(g)$ is the subprobability mass of j steps whose every partial sum is at least $-g$, then $p = q_0 \sum_{j \geq 0} a_j(g)$. At a candidate time with i steps before it and j after it, reverse and negate the past increments. Symmetry and independence give mass $q_0 a_i(g) a_j(g)$ for the candidate to lie within g of the minimum. Summing in i, j gives p^2/q_0 . Every qualifying candidate is counted; no unique minimum is selected.

Bounding the occupation count. The killed kernel is dominated by the original displacement law, including after convolution:

$$Q_S^{*k}(w) \leq \Pr(W_1 + \cdots + W_k = w).$$

Thus the atom estimate above bounds the expected number L_A of visits to any interval A of width $2g$, uniformly in the starting position, by

$$\mathbb{E}L_A \leq 1 + \sum_{k \geq 1} \min\{1, Cgk^{-1/b}\} \leq Cg^b.$$

Markov's inequality and restarting after successive visit-count thresholds give an exponential tail for L_A at scale g^b . Anchor an interval of radius g at each alive index in turn and count its future visits. The earliest near-minimum index captures all N_g near-minimum visits in its interval. We do not condition at that retrospectively chosen index: instead union bound over all indices, whose survival probabilities sum to $1/q_0$. Consequently

$$\Pr(N_g \geq t) \leq \min\{1, (C/q_0)e^{-ct/g^b}\}, \quad \mathbb{E}N_g \leq Cg^b \log(2 + 1/q_0).$$

Together with the factorization this bounds p by $C(g/S)^{b/2} \sqrt{\log(2 + S)}$.

Finally, preserving weak order below S also preserves it below E on $\{E < S\}$. Independence of E from the original sequences gives

$$(1 - e^{-1})\Pi_g(S) \leq p.$$

Since $b/2 = 1/4$, this proves the lemma. $\square$

9.4 Summing the marked-arm decomposition

Proof of Theorem 9.1. We now have the two different survival tests needed for an arch: the shared-turn test at its top and Lemma 9.4 at its bottom. Write $b = 1/2$, as in the proof of that lemma. Use the two-arm decomposition above, whose actual arms are disjoint after the turn. We will prove (9.1) by allocating independent groups of pieces to these tests, the reward, and the two-coordinate match.

Use dyads $H \leq h < 2H$. Those $H < g/\log^2 g$ contribute negligibly by exponential strip aspect since one bridge has diameter of order at least g . On remaining dyads diameters above $CH \log^2 H$ cost $O_A(H^{-A})$ for arbitrary fixed A , still by strip aspect (sum bridge products at each height, also with lengths using area). Thus in discards with superpolynomial accuracy we may pay a polynomial bound for lengths.

Let k_i be the two irreducible counts, in dyads at scales K_i , $K = \max K_i$. Treat their bridge masses at specified counts as product iid laws. Discard $k_i > H^{a+\eta}$ by height damping, with $\eta > 0$ arbitrarily small fixed ($k_i \leq 2H$). Expand the length weight by marking one piece on its arm. On the other arm also mark a large height jump $> H^{1-\eta/(2a)}$ (upper bound by counting choices), if $k_i \leq H^{a-\eta}$; without one the height requirement costs superpolynomially little, by exponential weighting at the inverse cutoff on the truncated jumps, using $\Pr(Y > t) \leq Ct^{-a}$. For larger count on that arm just designate a middle index instead.

Sort the index distances (counting from 1) of each designated piece from the top and bottom ends into dyadic scales I_i, J_i respectively. Thus $\max(I_i, J_i) \asymp K_i$. The sum of index choices and the associated length or large-jump costs have budget, on the two sides combined,

$$C_\eta H^{d_* - 2a + O(\eta)} \prod_{i=1}^2 \min(I_i, J_i).$$

Indeed for a given length piece use Lemma 9.3; the jump costs $CH^{-a+O(\eta)}$, and the number of choices on each side is at most a constant times its smaller distance scale. In the middle-index case $\min(I_i, J_i)H^{-a+\eta} \gtrsim 1$ already.

Put $I = \min I_i$ and $J = \min J_i$. The following deterministic allocation makes all uses of independence explicit. Let r_i be the designated index on arm i , whose total count is k_i , and write

$$u_i = r_i - 1, \quad v_i = k_i - r_i, \quad u = \min_i u_i, \quad v = \min_i v_i.$$

Take the first $\lfloor u/16 \rfloor$ pieces of each arm for the top test and the last $\lfloor v/16 \rfloor$ for the bottom test. Neither block contains a designated piece. They are disjoint since $u + v \leq k_i - 1$ for both arms. On an arm with maximal count $k_{\max}$, at least $15k_{\max}/16 - 1$ indices remain after these blocks and its designated piece are removed. Reserve the first $\lfloor k_{\max}/4 \rfloor$ of these unused indices in their original order for the planar atom; condition on all remaining unreserved draws. The reserved draws need not be consecutive: their displacements still add and they remain independent with the original law.

If $u < 16$, the top distance scale I is bounded, so we omit that test and bound its probability by one; this changes the displayed cost below only by a constant. The same holds for $v < 16$ and J . Otherwise the two testing blocks have orders I and J . Bounded $k_{\max}$ is absorbed into constants in the atom bound. This allocation therefore covers marks next to endpoints, unequal arm lengths, and both possible designations. The factors to be multiplied are summarized here; $p' = b/(2a) = 1/3$, $m_i = \min(I_i, J_i)$, and the displayed powers of H allow fixed multiples of η .

Piece group	Role	Upper cost
Top blocks on both arms	Shared-turn order	$I^{-1}H^{O(\eta)}$
Designated pieces and indices	Reward and other-arm mark	$H^{d_* - 2a + O(\eta)} m_1 m_2$
Reserved larger-arm indices	Two-coordinate match	$K^{-2/a}$
Bottom blocks on both arms	Separated-port weak order	$g^{b/2} J^{-p'} H^{O(\eta)}$

The second row includes the designated-index counting allowance. For a large-jump mark this is the actual sum of index choices. In the deterministic middle-index case its cost one is bounded by $m_i H^{-a+\eta} \gtrsim 1$ on that arm, so the same combined budget applies; the factor $H^{-a+\eta}$ is not a probability bound on that unconditioned piece. Choices of total counts contribute at most $CK_1 K_2$ separately.

With $S_I = I^{1/a} H^{-C_0 \eta}$ for fixed sufficiently large C_0 , this many pieces reach height S_I except with superpolynomial error by height damping if $S_I \geq 2$; similarly at $S_J = J^{1/a} H^{-C_0 \eta}$. Both thresholds are far below h . On valid arms we thus impose Proposition 9.2(ii) up to S_I (rounded down), by the shared-turn crosscut argument in [15, Section 7.3], with the common top-port convention stated in Proposition 9.2(ii), at probability cost $\leq C_\eta I^{-1} H^{O(\eta)}$, allowing roles. At the other end impose weak order at the common cuts below S_J at cost $\leq C g^{b/2} J^{-b/(2a)} H^{O(\eta)}$, since the two disjoint prefixes to a common cut are strip crosscuts. For thresholds < 2 the corresponding upper costs with enlarged $O(\eta)$ need no test. Reversed shapes are again iid by bridge symmetry, tested as if placed at the prescribed two bottom ports (actual relative placement upon imposing the increment equalities); these tests therefore do not consume information from the middle increments.

In addition the required exact height equality and transverse difference together cost $\leq CK^{-2/a}$ by the local estimate on (Y, Z) , reserving order K other draws of the larger count side independently of both tests and the marks (adjust constant for bounded K). Including at most $CK_1 K_2$ pairs of counts, all count-scale factors cancel: writing $p' = b/(2a) = 1/3$,

$$K_1 K_2 K^{-2/a} \prod_i \min(I_i, J_i) I^{-1} J^{-p'} \leq CK^{-2/a} \max(I_1, I_2) \max(J_1, J_2) J^{1-p'} \leq C.$$

Thus including polylogarithmic bin sums proves (9.1). Since $d_* - 2a < 0$, summation and free choice of η give total length mass at most $g^{1/12+o(1)}$ for both arms.

The length-weighted bound is $K_g \mathbb{E}L \leq g^{1/12+o(1)}$. Dividing by $K_g \geq cg^{-5/4}$ proves the mean upper bound and the long-length probability bound by Markov's inequality. For the lower probability bound, every arch has diameter at least a fixed multiple of g . Sum Proposition 9.2(iii) over dyadic diameter bins starting there, taking a slightly smaller exponent slack to absorb unit and endpoint conventions. The resulting mass is superpolynomially small compared with K_g . Thus the short-length probability tends to zero; it also gives $\mathbb{E}L \geq g^{4/3-o(1)}$. This proves Theorem 9.1. $\square$

The following terminal-summed consequence also appears in the proof of [15, Theorem 8.1]. We include its normalization and first-moment argument to distinguish this confined mixture from the individual-gap law.

Corollary 9.5 (Confined macroscopic-gap arches). *Fix a half-plane boundary source and choose a sufficiently large fixed C . Give each arch with terminal gap in $[h, 2h]$ on one chosen side and diameter at most Ch its critical weight, then normalize over this entire set of terminal ports. For every integer $h \rightarrow \infty$,*

$$L(\gamma) = h^{4/3+o(1)} \quad \text{in probability,} \quad \mathbb{E}L(\gamma) = h^{4/3+o(1)}.$$

Proof. The confined pointwise arch lower bound from the calibrated finite input above, summed over the order h allowed gaps, gives normalizing mass at least $ch^{-1/4}$ once C is fixed sufficiently large. For the first moment, translate the source to order h distinct ports on the same straight side. All translated confined arches fit in one convex pure box of diameter $O_C(h)$. Distinct sources distinguish the translated paths, so the calibrated convex-domain length bound [12, Proposition 11.1], divided by the number of sources, gives first length mass at most $C_C h^{13/12}$. Division by the normalizer proves $\mathbb{E}L \leq C_C h^{4/3}$ and the upper probability tail by Markov's inequality. Every retained arch has diameter at least a fixed multiple of h . Proposition 9.2(iii), summed over the finitely many diameter bins through Ch , makes its short-length mass superpolynomially small. This proves the lower probability tail and hence the mean lower bound. The law in this corollary sums over terminal ports before normalization; it is distinct from the individual-gap law of Theorem 9.1. $\square$

10 Endpoint means from polynomial vacuum estimates

We now estimate mean lengths for bridges to prescribed top ports. The polynomial-vacuum representation controls endpoint variation after summing over heights. This gives individual-height estimates on one density-one set, and estimates at every large scale after critical averaging over even heights.

Let $\mathcal{B}_h(e)$ be the strict bridges from bottom coordinate 0 to top coordinate $e \in \mathbb{Z} + h/2$ at height h . Define

$$b_h(e) = \sum_{\gamma \in \mathcal{B}_h(e)} \kappa^{L(\gamma)}, \quad b_h^{(L)}(e) = \sum_{\gamma \in \mathcal{B}_h(e)} L(\gamma) \kappa^{L(\gamma)}.$$

Thus $b_h = \sum_e b_h(e) = B_h$ and $b_0(e) = \mathbf{1}_{e=0}$.

Theorem 10.1 (One set of density-one heights). *There is one set $G \subseteq \mathbb{N}$ with $|G \cap [1, N]|/N \rightarrow 1$ such that, for every fixed $\delta, \varepsilon > 0$, all sufficiently large $h \in G$ satisfy*

$$h^{4/3-\varepsilon} \leq \frac{b_h^{(L)}(e)}{b_h(e)} \leq h^{4/3+\varepsilon} \quad (e \in \mathbb{Z} + h/2, |e| \leq h^{1-\delta}).$$

For $E_H = 2\mathbb{N} \cap [H, 2H]$, put $Z_H = \sum_{h \in E_H} b_h(0)$ and $M_H = \sum_{h \in E_H} b_h^{(L)}(0)$.

Theorem 10.2 (All-even aligned height mixture). *Give each pair (h, γ) , $h \in E_H$ and $\gamma \in \mathcal{B}_h(0)$, probability $\kappa^{L(\gamma)}/Z_H$. As $H \rightarrow \infty$ through all real scales,*

$$Z_H = H^{-1/4+o(1)}, \quad M_H = H^{13/12+o(1)}, \quad \mathbb{E}_H L = M_H/Z_H = H^{4/3+o(1)}.$$

The height marginal is $b_h(0)/Z_H$.

The complete finite construction in [14, Theorem 1.1 and Lemma 8.2] gives $b_h = h^{-1/4+o(1)}$ and $\sum_e b_h^{(L)}(e) = h^{13/12+o(1)}$, with exponential strip aspect cutoff. The flat-strip flux identity in [14, Section 2] is $b_h + \cos(3\pi/8)A_h = 1$, where A_h is the raw critical mass of nontrivial paths that stay inside the height- h strip, from the fixed bottom source back to the bottom wall, with the terminal port summed. Let i be its positive-height irreducible probability, now recording height and transverse displacement. Its array $i^{(L)}(h, e)$ is the sum of $L(\gamma)\kappa^{L(\gamma)}$ over irreducibles with those height and displacement coordinates. Convolution below is in both variables. We use the following exact reward input from [15, Proposition 10.1].

Proposition 10.3 (Polynomial-vacuum rewards). *Write J for the height of one piece under i , and L for its length. Set $\alpha = 1/4$, $\beta = 3/4$, and $d_* = 4/3$. Then*

$$b = \sum_{k \geq 0} i^{*k}, \quad b^{(L)} = b * i^{(L)} * b,$$

$$i(J > r) = r^{-\beta+o(1)}, \quad \mathbb{E}_i[L; J \leq r] \leq r^{d_*-\beta+o(1)}, \quad \mathbb{E}_i[Le^{-J/r}] = r^{d_*-\beta+o(1)}.$$

The discounted lower reward is retained on pieces with height at most $r \log^2 r$ and lateral excursion at most $r \log^4 r$.

Only in this section, J denotes height. We use $\sigma = 3/8$ in the winding phases and $\theta = \pi/8$ in their coefficients.

Reflection makes $b_h(e)$ symmetric in e . Heights and offsets both add under concatenation, so the two convolution identities apply to these arrays. The lower estimates below will hold at every large height. The endpoint upper estimates will instead be summed over heights; this distinction is what leads to the two different conclusions of the section.

10.1 Endpoint variation from two disjoint bridges

First put t_h equal to the total critical product mass for two vertex-disjoint bridges with adjacent starts 0,1 in that order. Write $J_h(u, v)$ for its refinement by ordered endpoints $u < v$ on top. We claim

$$b_h(y) = \sum_{u \leq y < v} J_h(u, v).$$

Indeed take a bridge spectator from 1 to y' on top and grow an active exit from 0 with phase $e^{i\sigma W}$, stopping instead just before first contact with the spectator vertices. Take also the balance reversing the roles of 0 and 1 (spectator now from 0 to y'), multiplied by $s_0 = e^{6i\theta}$, $\theta = \pi/8$. Revisit pairs cancel; convergence at fixed width in the vertical coordinate (the unbounded horizontal strip) follows by the finite strip exponential lateral cutoff.

At a first contact the three arms from 0, 1, y' form a tripod with the central weight included once, with bijections from the contact terms of each balance. The three branches have the same

turn phases at the first contact, so their turn phases measured before turning at the center satisfy $e^{i\sigma W_{y'}} = e^{i\theta} e^{i\sigma W_0} = e^{-i\theta} e^{i\sigma W_1}$. Hence the contact terms cancel in this combination ($s_0 = -e^{-2i\theta}$).

The remaining active arches in the first balance can only go left, separated by the bridge from exits beyond the adjacent port (also no exit at it); in the second only right. Their phases including the combination factor are thus both $\ell = e^{3i\theta}$. Project by $X \mapsto \Im(\ell^{-1}X)/\Im(\ell^{-1})$, killing them and giving coefficients 1, -1 on the two types of bridge terms. This yields

$$b_h(y' - 1) - b_h(y') = \sum_{u < y'} J_h(u, y') - \sum_{v > y'} J_h(y', v).$$

Sum up to $y' = y$.

Consequently $\sup_e b_h(e) \leq t_h$ and the sum of absolute first differences of $b_h(\cdot)$ on its coset is at most $2t_h$. Also $t_h \leq C(A_{h+1} - A_h)$: reflect so the adjacent ports are on the top of the old strip, join them by the bounded minimal arch in the added one layer, and translate horizontally to put the resulting bottom arch's starting port at the fixed source (use the ordered ends). A strip-exchanging lattice isometry with any needed offset can be used before this translation. Recovery has bounded multiplicity since the crossing ports of the old top line are exactly the two joined ports, adjacent and prescribed before translation. Thus by flat strip flux, for fixed $0 < c_1 < c_2 < \infty$,

$$\sum_{c_1 H \leq j \leq c_2 H} t_j \leq C b_{\lfloor c_1 H \rfloor} = H^{-\alpha+o(1)} \quad (\text{as an upper bound}).$$

10.2 A pointwise lower bound at every height

The variation bound is available only after summing heights. Nevertheless it will yield a lower bound at each height by averaging internal renewal cuts. Here is a uniform pointwise **lower** bound at all large heights for any fixed $\delta > 0$:

$$b_h(e) \geq h^{-1-\alpha-o(1)} \quad (|e| \leq h^{1-\delta}, e \in \mathbb{Z} + h/2).$$

First take even h . For a bridge $\gamma \in \mathcal{B}_h(e)$, let $N_h(\gamma)$ count the integers $j \in [h/8, h/7]$ for which all three levels $j, 2j, h/2 + j$ are single-crossing cuts. Put $k = h/2 - j$. Splitting at those cuts gives four bridges of heights j, j, k, k , so, with convolution here only in the offset,

$$\sum_{\gamma \in \mathcal{B}_h(e)} N_h(\gamma) \kappa^L(\gamma) = \sum_{h/8 \leq j \leq h/7} (b_j * b_j * b_k * b_k)(e).$$

For all but $o(h)$ of these internal levels j , the summed variation estimate gives $t_j \leq h^{-1-\alpha+\delta_0/8}$, where $\delta_0 = \min(\delta, 1/2)$. The external height h remains arbitrary. Write $A = b_j(\cdot) * b_k(\cdot)$. Its total is $b_j b_k$, concentrated on $|x| \leq Ch \log^2 h$ up to negligible tails by the exponential horizontal strip bounds; its sup norm and total variation on its coset are bounded by $t_j b_k, 2t_j b_k$ respectively. Thus

$$\sum_x A(x) A(e - x) = \sum_x A(x) A(x - e) \geq \|A\|_2^2 - 2|e| t_j^2 b_k^2, \quad \|A\|_2^2 \geq h^{-1-o(1)} (b_j b_k)^2.$$

For good j the subtracted term is negligible. Summing gives first moment at least $h^{-4\alpha-o(1)}$.

For the second moment, diagonal terms cost at most $h^{-4\alpha+o(1)}$ in sum (use $\sup b_j(\cdot) \leq t_j$ on one factor). Off-diagonal terms $j < j'$ are cut up by the union of both triples of levels, yielding endpoint convolution masses of seven bridges: four of heights comparable to h , including j , and

three of heights $\Delta, 2\Delta, \Delta$, $\Delta = j' - j$. Bound pointwise using the sup at j and total masses for all others. Since $\sum_{\Delta \leq h} b_\Delta^2 b_{2\Delta} \leq h^{1-3\alpha+o(1)}$, summation of t_j gives

$$\sum_{\gamma \in \mathcal{B}_h(e)} N_h(\gamma)^2 \kappa^{L(\gamma)} \leq h^{1-7\alpha+o(1)}.$$

Weighted Cauchy–Schwarz therefore gives

$$b_h(e) \geq \frac{(\sum_{\gamma \in \mathcal{B}_h(e)} N_h(\gamma) \kappa^{L(\gamma)})^2}{\sum_{\gamma \in \mathcal{B}_h(e)} N_h(\gamma)^2 \kappa^{L(\gamma)}} \geq h^{-1-\alpha-o(1)}.$$

Odd heights follow by a bounded one-layer extension from even height with the appropriate half-step endpoint shift, using a lower-bound window with a smaller positive δ .

10.3 Inserting length between two bridge kernels

Write $K = b * b$ in both height and offset. It follows that

$$K_h(e) \geq h^{-2\alpha-o(1)} \quad (|e| \leq h^{1-\delta})$$

again for admissible offsets. Indeed sum over heights $h/3 \leq l \leq 2h/3$ and offsets $|x| \leq h^{1-\eta}$ of the first factor, for arbitrarily small $0 < \eta < \delta/2$; both factors use the preceding lower bound with sublinear power windows. Send η down. On the upper side we have

$$\sum_{j \leq 2H} \sup_x K_j(x) \leq H^{2\beta-1+o(1)}.$$

Order the two factor heights by size, larger l and smaller r . Apart from zero total height use $\sup b_l(\cdot) \leq t_l$ and $\sum_{r \leq l} b_r \leq l^{\beta+o(1)}$ (up to constants); summing dyadically gives the bound since $\beta - \alpha > 0$.

The length insertion is exactly $b^{(L)} = K * i^{(L)}$. Uniformly in the same windows we obtain the pointwise lower bound

$$b_h^{(L)}(e) \geq h^{d_*+\beta-2-o(1)}.$$

In fact take the middle piece from $i^{(L)}$ with height $\leq m \log^2 m$ and absolute displacement $\leq m \log^4 m$, $m = h^{1-\varepsilon}$, for small fixed $\varepsilon > 0$. Its available total mass is at least $m^{d_*-\beta-o(1)}$, by the exponentially height-discounted first moment and tail truncations in Proposition 10.3. The remaining K uses height comparable to h . Since $m \log^4 m \leq h^{1-\varepsilon/2}$ for large h , its offset still lies in a sublinear power window for each fixed $\delta, \varepsilon > 0$. Use the lower bound for K and send ε to zero. Also

$$\sum_{H \leq h \leq 2H} \sup_e b_h^{(L)}(e) \leq H^{d_*+\beta-1+o(1)}$$

by the upper K estimate and the truncated $i^{(L)}$ upper bound up to height $2H$.

10.4 One density-one set and the aligned mixture

Put $U_h = \sup_e b_h(e)$ and $V_h = \sup_e b_h^{(L)}(e)$, with the suprema over the actual row coset. For dyadic H , choose $\rho_H \rightarrow \infty$ with $\rho_H = H^{o(1)}$ so that the two summed upper bounds give

$$\sum_{H \leq h < 2H} U_h \leq \rho_H H^{-\alpha}, \quad \sum_{H \leq h < 2H} V_h \leq \rho_H H^{d_*+\beta-1}.$$

In each such interval retain precisely the heights satisfying

$$U_h \leq \rho_H^2 H^{-1-\alpha}, \quad V_h \leq \rho_H^2 H^{d_*+\beta-2},$$

and let G be the union of the retained heights. At most $2H/\rho_H = o(H)$ heights are removed from each interval. Summing over dyadic intervals shows that G has density one. On G ,

$$\sup_e b_h(e) \leq h^{-5/4+o(1)}, \quad \sup_e b_h^{(L)}(e) \leq h^{1/12+o(1)}.$$

The construction of G uses global endpoint suprema and does not depend on δ or ε . For each fixed $\delta > 0$, the all-height lower bounds give, uniformly on its admissible window, $b_h(e) \geq h^{-5/4-o(1)}$ and $b_h^{(L)}(e) \geq h^{1/12-o(1)}$. Combining the lower numerator with the upper denominator, and then the upper numerator with the lower denominator, yields

$$h^{4/3-o(1)} \leq \frac{b_h^{(L)}(e)}{b_h(e)} \leq h^{4/3+o(1)} \quad (h \in G, |e| \leq h^{1-\delta}).$$

This proves Theorem 10.1 with its stated quantifiers.

For completeness, the mixture in Theorem 10.2 follows directly from the estimates before that deletion of heights. At each large even height, the aligned lower bounds are $b_h(0) \geq h^{-5/4-o(1)}$ and $b_h^{(L)}(0) \geq h^{1/12-o(1)}$. Summing over the order H even heights in $[H, 2H]$ gives $Z_H \geq H^{-1/4-o(1)}$ and $M_H \geq H^{13/12-o(1)}$. The dyadic sums of the endpoint suprema give the reverse bounds $Z_H \leq H^{-1/4+o(1)}$ and $M_H \leq H^{13/12+o(1)}$. Their ratio proves the mixture mean at every large scale. No upper estimate at an individual discarded height has been used.

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